Reading skill and the time course of morphological processing: electrophysiological evidence for stage-specific modulation

Joanna Morris [1,2,*], **Emma Kealey** [2,3], **Frankie Greene** [1,4], **Joemari Pulido** [2,5], **Tess Rooney** [2,6], **Ethan Moore** [2], **Crismar Ramos-Marte** [2], **Natalia Alzate** [2]

[1] *School of Cognitive Science, Hampshire College, Amherst, MA, USA*
[2] *Department of Psychology, Providence College, Providence, RI, USA*
[3] *Department of Linguistics, University of Southern California, Los Angeles, CA, USA*
[4] *Oregon Health & Science University and Portland State University School of Public Health, Portland, OR, USA*
[5] *Department of Cognitive, Linguistic & Psychological Sciences, Brown University, Providence, RI, USA*
[6] *Columbia University School of Nursing, New York, NY, USA*

Correspondence*:
Joanna Morris
jmorris6@providence.edu

ABSTRACT

Skilled reading of morphologically complex words depends on both high-quality lexical representations and sensitivity to morphological structure. However, we do not yet understand how these abilities shape the time course of morphological processing. In this study, event-related potentials (ERPs) and reaction times were recorded during a lexical decision task on morphologically complex words and nonwords; both varied in base frequency and family size, and nonwords additionally varied in morphological complexity. Individual differences were quantified via principal components analysis of vocabulary knowledge, spelling ability, and print exposure, yielding two orthogonal dimensions: Language Proficiency (LP) and Orthographic Sensitivity (OS). Behaviourally, response times were sensitive to base frequency and family size for words, and to morphological complexity for nonwords, while individual differences played a minor role: Orthographic Sensitivity showed a modest effect on response speed for nonwords only, and Language Proficiency showed no reliable effect. ERP measures revealed a richer picture. For familiar words, Orthographic Sensitivity primarily modulated the N250 base frequency effect, with a smaller Language Proficiency effect too; only Language Proficiency modulated the N400 base frequency effect. For novel morphologically complex forms, both Orthographic Sensitivity and Language Proficiency modulated the cost of morphological complexity at the N250. For both dimensions, the complexity cost shrank as skill increased and reversed direction at the highest

skill levels. Because both dimensions were involved, this pattern qualifies the prediction that Orthographic Sensitivity alone governs early morphological parsing: instead, early parsing of novel forms appears to draw on reading skill more broadly. At the N400, a large morphological family made complex nonwords harder to process for readers lower in Language Proficiency, but this cost was absent for readers higher in Language Proficiency; Orthographic Sensitivity showed no comparable effect, giving the clearest evidence for the predicted dissociation between the two dimensions. Together these findings indicate that individual differences in reading skill are expressed chiefly in the dynamics of neural processing rather than in overt response times, and that the dissociation between orthographic and lexical-semantic profiles is clearest at the later stage where word meaning is accessed, while early morphological parsing of novel forms depends on both dimensions.

Keywords: morphological processing, individual differences, event-related potentials, N250, N400, lexical decision

1 INTRODUCTION

Reading is often described as "the process of gaining meaning from print" (Rayner et al., 2001, p. 34). Although comprehension ultimately depends on sentence- and discourse-level integration, efficient access to individual word forms and meanings remains a foundational component of skilled reading. However, readers vary considerably in this ability (Perfetti, 2007; Castles and Nation, 2006; Stanovich and West, 1989). Understanding what drives this variation requires identifying variables that are selectively diagnostic of the mechanisms underlying word recognition. Two such variables, base frequency (also referred to as stem frequency in the literature) and morphological family size, have proven especially informative for this purpose (Taft, 2004, 1979; de Jong et al., 2000; Bertram et al., 2000). Base frequency, the cumulative token frequency of a base morpheme across all its surface forms, and morphological family size, the number of distinct words sharing that base, both facilitate visual word recognition, yet their effects are independent and have been argued to reflect different levels of morphological representation: base frequency has been argued to index form-based access, reflecting the cumulative ease with which a base morpheme's orthographic form is processed across its surface variants, while family size has been argued to index the richness of semantic representation, reflecting the breadth of meaning-level connections activated when a morphological base is encountered (Schreuder and Baayen, 1997). The present study exploits this dissociation to ask how readers who differ in the precision of their orthographic and semantic knowledge draw differentially on form-based and meaning-based information during morphological processing.

Both variables are properties of a word's morphological family, and morphological structure plays a particularly important role in the English writing system. English exhibits irregular and context-sensitive grapheme–phoneme mappings, rendering phonological decoding alone an insufficient basis for fluent word recognition (Ziegler and Goswami, 2005; Castles and Nation, 2006). At the same time, English preserves regularities at higher representational levels, particularly in its morphological structure; words that share a stem or affix often exhibit consistent orthographic patterns and systematic semantic relationships (e.g., *heal–health* share a stem; *teacher*, *baker*, and *driver* share the agentive suffix *-er*). Skilled reading in English therefore requires sensitivity not only to phonological correspondences but also to recurring morphological structure that links form and meaning, and readers differ in how deeply and precisely that structure is represented (Bryant et al., 2005; Arciuli, 2018; Treiman, 2018).

Base frequency and family size are the primary variables used to study this structural sensitivity; before characterising how readers differ in processing them, however, it is necessary to clarify what each variable indexes and how it has been theoretically interpreted. This question has proven theoretically contentious because the answer depends on the architecture one ascribes to the word recognition system. Whether skilled readers decompose complex words into constituent morphemes, access whole-word representations directly, or exploit learned statistical regularities carries different implications for what each variable measures, that is, whether it reflects the strength of a stored morpheme entry, the output of a parsing procedure, or an emergent property of a statistical learning system. As a result, what each variable measures determines what kind of reader variation should modulate its effect. The Lexical Quality Hypothesis (Perfetti and Hart, 2002; Perfetti, 2007, 2017) provides a principled framework for characterising that variation: lexical representations vary across readers in how completely and precisely a word's orthographic and semantic constituents are specified, and it is this variation in representational quality that determines how strongly each variable influences processing. Although competing architectures disagree about the mechanism responsible, they converge in treating base frequency as an index of how precisely the orthographic form of the base morpheme is specified, and family size as an index of how richly and flexibly its meaning is represented across morphological contexts. Readers who differ in orthographic precision versus semantic breadth should therefore differ selectively in their sensitivity to each variable.

1.1 Base Frequency, Morphological Family Size, and Competing Theoretical Accounts

Both base frequency and morphological family size facilitate visual word recognition, and their effects are independent when appropriately controlled (Ford et al., 2010). The theoretical significance of this dissociation is that it constrains what any model of morphological processing must explain: current accounts converge on the prediction that base frequency effects arise earlier in processing than family size effects, but diverge sharply in the mechanisms responsible.

Dual-route and multi-stage accounts agree that the two variables index distinct levels of processing, but disagree on the nature of those levels. In classical dual-route frameworks (Schreuder and Baayen, 1995; de Jong et al., 2000), base frequency reflects the ease of accessing a stored morpheme entry via a decompositional route, while family size reflects the richness of the semantic network surrounding the base, attributed to the whole-form route. Early automatic decomposition accounts (Taft and Forster, 1975; Taft, 2004) reconceptualise this dissociation in temporal rather than architectural terms: decomposition is not one route among several but an obligatory initial parsing step that precedes semantic access. On the morpho-orthographic formulation of this view (Rastle and Davis, 2008; Longtin and Meunier, 2005; Morris et al., 2007), an intermediate level of representation mediates between sublexical letter clusters and whole-word forms, and base frequency reflects the strength of activation at this stage. Family size effects are then linked to later semantic processing, consistent with the finding that opaque relatives contribute less to the family size effect than transparent ones (Schreuder and Baayen, 1997; de Jong et al., 2000).

The discriminative learning framework (NDL/LDL; Chuang and Baayen, 2021) offers a more radical reinterpretation. On this account, lexical processing is modelled as learned form-to-meaning mappings driven by prediction error rather than by decompositional morpheme retrieval. Base frequency reflects the accumulated strength of the n-gram cues that spell out the base across all its surface forms, while family size reflects the type diversity of the training signal; a larger family causes shared cues to converge on a more robustly specified semantic core. Both effects arise from the same learning mechanism, rendering the dissociation between them a consequence of different statistical properties of the input rather than evidence for distinct representational levels or processing routes.

Taken together, these accounts agree that base frequency and family size exert independent effects, with base frequency arising earlier in processing than family size. Where they diverge is in the nature of the representations involved—symbolic morpheme entries, learned orthographic patterns, or weighted form-to-meaning mappings—and consequently in how they explain the source of individual differences in sensitivity to each variable. Dual-route and morpho-orthographic accounts imply that readers with more precise sublexical orthographic representations should show greater sensitivity to base frequency, since this variable indexes activation strength at the form-parsing stage, while readers with richer semantic networks should show greater sensitivity to family size, since this variable indexes the breadth of meaning-level connectivity. The discriminative learning framework leads to a similar expectation but for different reasons: individual differences in statistical learning history determine the specificity of both orthographic cue weights and semantic vectors, so readers with more extensive print exposure should show stronger and more differentiated sensitivity to both variables. However, all three accounts are consistent with the view that form-level and meaning-level sensitivity should be dissociable, and that this dissociation should be traceable to distinct processing stages—early for base frequency, later for family size.

Across these accounts, a recurring question is whether morphological decomposition is driven primarily by affix recognition or by stem activation. Affix-stripping accounts (Taft and Forster, 1975; Rastle and Davis, 2008) propose that the affix is identified first and triggers decomposition, after which the stem is looked up in the lexicon. However, Beyersmann et al. (2016) found that high proficiency readers showed embedded stem priming effects independent of affix status: priming occurred even when the stem was followed by a non-affixal ending, suggesting that stem activation in skilled readers is not contingent on affix recognition. This finding suggests that affix-stripping alone may be insufficient to account for morphological processing in skilled readers, and raises the possibility that stem- and affix-based mechanisms operate concurrently, with their relative contributions varying as a function of individual differences in reading proficiency. The present study tests this possibility directly by examining whether Orthographic Sensitivity and Language Proficiency, two dissociable dimensions of individual variation in reading skill, differentially modulate early morpho-orthographic parsing in ways consistent with affix-based and stem-based mechanisms respectively.

1.2 Individual Differences in Morphological Processing

Traditional models of word recognition have typically characterised processing at the level of the skilled reader in general, abstracting over individual differences in favour of a uniform cognitive architecture. Yet readers vary considerably in how they process written words, and understanding this variation offers a window into the representations and processes underlying skilled reading. The Lexical Quality Hypothesis, introduced above, provides a principled framework for understanding this variation. Readers do not differ along a single dimension of lexical quality. Andrews and colleagues (Andrews, 2012; Andrews and Lo, 2013) have demonstrated that individual differences in orthographic and semantic knowledge are dissociable and predict distinct patterns of morphological processing. Readers with a semantic profile, defined by relatively higher vocabulary than spelling ability, show robust priming for transparent morphological pairs but little priming for opaque or form-only pairs, particularly for slower responses. Readers with an orthographic profile, defined by relatively higher spelling than vocabulary, show sustained priming for opaque pairs that is at least as strong as for transparent pairs. Andrews and Lo (2013) interpret this pattern as evidence that a systematic component of individual differences among skilled readers is explicable in terms of the balance between orthographic and semantic processing; readers differ not merely in overall processing efficiency but in the relative weight they assign to form-based versus meaning-based information during lexical access.

The present study builds directly on this finding. If base frequency and family size index form-based and meaning-based aspects of lexical representation respectively, then readers who differ in their balance of orthographic and semantic processing should differ predictably in their sensitivity to each variable. Moreover, if these differences have a temporal structure, with form-based processing preceding semantic processing in the recognition cascade, they should be traceable to distinct ERP components. To capture the full range of this variation continuously rather than as a categorical split, the present study derives two orthogonal individual difference dimensions, Language Proficiency and Orthographic Sensitivity, from a principal components analysis of vocabulary knowledge, spelling ability, and print exposure, following the approach of Andrews and Lo (2013). If base frequency and family size are differentially modulated by these two dimensions, and if those modulations emerge at distinct stages in the ERP, this would constitute direct electrophysiological evidence that orthographic and semantic processing represent a genuine and temporally structured axis of variation in the human reading system.

1.3 Neural Indices of Morphological Processing

To track whether base frequency and family size effects emerge at distinct temporal stages, and whether individual differences in orthographic and semantic processing modulate those stages selectively, the present study uses two well-established ERP components as dependent measures—the N250, as an index of sublexical processing, and the N400, as an index of lexical-semantic integration. These components have been argued to reflect sequential stages in the cascade from orthographic decoding to meaning retrieval during visual word recognition (Grainger and Holcomb, 2009).

The N250 is a negative-going deflection peaking at approximately 250 ms post-stimulus onset. Holcomb and Grainger (2006) propose that it reflects the stage at which abstract sublexical orthographic representations, that is letters and letter clusters, are mapped onto higher-level form representations prior to full lexical access. Evidence for this interpretation comes from its graded sensitivity to orthographic overlap between prime and target in masked priming; N250 amplitude is most attenuated for full identity repetitions, intermediate for primes sharing all but one letter, and largest for fully unrelated pairs (Holcomb and Grainger, 2006, 2007). The N250 is also sensitive to sublexical phonological overlap, with Grainger et al. (2006) identifying separable early orthographic and later phonological subcomponents, consistent with dual-route architectures of word recognition.

The N400 is a negative-going component peaking around 400 ms, with a broad centro-parietal distribution (Kutas and Hillyard, 1980). Across more than four decades of research, its amplitude has been shown to vary with semantic priming, cloze probability, lexical association, concreteness, and semantic richness, while remaining insensitive to physical and syntactic manipulations that leave meaning intact (Kutas and Federmeier, 2011). For the present study, the N400 is reliably modulated in single-word and word-pair paradigms, with larger amplitudes for semantically unrelated than related targets, indexing the ease with which a word's meaning is integrated with concurrently activated semantic information: in the present design, the network of meanings associated with a stem's morphological family, rather than sentence- or discourse-level content.

Although both components are negative-going deflections occurring in adjacent time windows, they are dissociable by topography—the N250 has a more frontal distribution, the N400 a posterior one (Holcomb and Grainger, 2006)—and, more decisively, by a double dissociation across priming types. Lexical status selectively modulates the N400 in that priming effects are restricted to real words with semantic representations and do not extend to pseudowords (Kiyonaga et al., 2007), whereas the N250 is equally modulated by words and pseudowords alike. The reverse pattern holds for semantic relatedness; while associative priming reliably modulates the N400, it produces no N250 effect (Grainger and Holcomb, 2009). This double dissociation—pseudoword priming engaging the N250 but not the N400, semantic priming engaging the N400 but not the N250—confirms that the two components index qualitatively distinct processing stages, and rules out the interpretation of the N250 as merely an early onset of the N400.

1.4 Aims and Predictions

The present study has two complementary aims. The first is to establish whether base frequency and morphological family size exert their effects at distinct temporal stages of visual word recognition, using ERP to track processing in real time. The second, and primary, aim builds on this: to examine whether individual differences in orthographic and semantic processing, captured by two continuous dimensions, Orthographic Sensitivity (OS) and Language Proficiency (LP), derived from a principal components analysis of spelling ability, vocabulary knowledge, and print exposure, modulate these effects selectively, at the processing stages where each variable is predicted to operate.

Although OS and LP are orthogonal dimensions rather than poles of a single continuum, they nonetheless differ in their primary loading: OS captures relatively greater precision of sublexical orthographic representation, while LP captures relatively greater breadth of lexical-semantic knowledge. The core prediction is therefore a pattern of component-specific modulation: Orthographic Sensitivity should modulate base frequency effects at the level of early form-based processing, as indexed by the N250, while Language Proficiency should modulate family size effects at the level of semantic integration, as indexed by the N400. This prediction follows directly from Andrews and Lo's (2013) finding that the balance between orthographic and semantic processing is a genuine and measurable axis of individual variation, combined with the broadly shared theoretical assumption that base frequency reflects form-level access and family size reflects the richness of semantic representation. If this pattern of component-specific modulation is observed, it would constitute direct electrophysiological evidence that the distinction between form-based and meaning-based morphological processing is not merely a property of competing computational models but a real axis of variation in the reading system. If, instead, both variables are modulated by the same individual difference dimension at the same processing stage, this would suggest that the two effects are less dissociable than current multi-stage accounts propose.

The first analysis examines these predictions in real word recognition. Because real words were not matched across morphological variables, reflecting the naturalistic distributional properties of morphologically complex words rather than an orthogonally controlled factorial design, base frequency, family size, and morphological complexity are treated as continuous predictors in a regression framework for reaction times. For ERP data, items were median-split into high and low groups separately for base frequency and family size to yield stable waveforms for comparison. The two measures therefore carry somewhat different inferential weight; the RT regression speaks to the independent continuous contributions of each variable, while the ERP analyses speak to the direction and timing of effects at the extremes of each variable's distribution.

The second analysis examines whether these individual difference modulations generalise to novel morphological structure, using morphologically complex nonwords. Morphologically structured nonwords (e.g., *gasful*: real stem + real affix) were compared against matched controls (e.g., *gasfil*: real stem + pseudoaffix), holding stem identity constant while manipulating suffix legitimacy. Because these items have no stored whole-word representations, they place maximal pressure on the system's generalisation mechanism and allow stem-level variables to be examined independently of whole-word familiarity. The key prediction is that Orthographic Sensitivity should modulate early N250 responses to the suffix contrast. For nonwords specifically, no stored whole-word entry exists, so early processing must rely on readers' sublexical orthographic knowledge. Readers with more precise representations of real suffix forms should show greater N250 sensitivity to the distinction between real and pseudo-suffixes, consistent with affix-led parsing accounts (Taft and Forster, 1975; Rastle and Davis, 2008). Language Proficiency, by contrast, should predict later N400 modulation; readers with broader and more richly specified lexical-semantic knowledge should show stronger N400 sensitivity to whether a nonword's stem belongs to a large or productive morphological family, since it is this family membership that determines the richness of the semantic activation the form can support. However, the predicted dissociation between OS at the N250 and LP at the N400 may not be fully clean for nonwords. Beyersmann et al. (2016) found that readers higher in overall reading proficiency showed embedded stem priming effects independent of affix status, suggesting that stem activation in skilled readers is not contingent on affix recognition and that stem- and affix-based parsing mechanisms may operate concurrently. Because Beyersmann et al.'s proficiency measure was not decomposed into orthographic and lexical-semantic components, it is unclear whether this effect reflects Orthographic Sensitivity, Language Proficiency, or both. The present study is therefore in a position to address this question more precisely: if stem-based parsing is specifically associated with Language Proficiency rather than Orthographic Sensitivity, this would suggest that the stem-based mechanism is driven by depth of lexical-semantic knowledge rather than sublexical orthographic sensitivity, consistent with the theoretical distinction between the two dimensions developed above. This leads to the prediction that LP may contribute to N250 modulation for nonwords alongside OS, potentially through a stem-based parsing mechanism that is distinct from the affix-led account. This is an interpretation that the complex versus simple nonword contrast is well-suited to evaluate, since it holds stem identity constant while manipulating suffix legitimacy.

2 METHODS

2.1 Materials

All stimulus items—200 nonword words, 100 morphologically complex words, and 100 monomorphemic filler items—were adapted from Sawi (2016). The 200 nonwords, were constructed to manipulate morphological structure while controlling surface form. In the *complex nonword* condition, existing English base words (e.g., *gas*) were combined with existing derivational suffixes (e.g., *gasful*), to yield 100 morphologically plausible complex nonwords. Each combination was syntactically legal and reflected common English derivational patterns. A total of 16 suffixes were used, each attached to four different stems.

In the *simple nonword* condition, the same base words as in the complex non-word condition were combined with non-suffix endings that were orthographically similar to the real suffixes used in the complex condition but did not constitute valid morphemes, to yield 100 orthographically plausible simple non-words. These pseudoaffix endings were created by changing the central letter of each suffix (e.g., *gasful–gasfil*), preserving length and orthographic neighborhood characteristics while eliminating morphological structure. Thus each complex non-word had an orthographically matched simple counterpart.

Because the same base appeared in both the simple and complex conditions, the experimental nonwords were distributed across two lists such that no participant encountered the same base more than once. In addition to the experimental nonwords, the stimulus set included 100 monomorphemic fillers (50 real words and 50 nonwords) to balance lexical status and reduce predictability. The experimental stimuli were drawn from Sawi (2016). Estimates of cumulative base (root) frequency, surface frequency, and morphological family size were obtained from the CELEX lexical database. Morphological family size was calculated by identifying all morphologically related forms sharing the same base (e.g., *calculate*, *calculable*, *calculation*, *calculator*). Following prior work (e.g., de Jong et al., 2000, 2002), both compounds (e.g., *WATCHTOWER*) and hyphenated compounds (e.g., *CHECK-IN*) were included, as they contribute to morphological connectivity within the lexicon. Base frequency was computed by summing the lemma frequencies of all confirmed family members.

The morphologically complex word stimuli consisted of 100 semantically transparent words with exclusively productive suffixes, a subset of the item set from Ford et al. (2010) selected by Sawi (2016). The words varied in base and surface frequency as well as in morphological family size.

For reaction time analyses, all lexical and individual-difference variables were treated as continuous predictors to preserve information and maximize statistical power. In contrast, ERP analyses were conducted on condition-averaged waveforms to improve signal-to-noise ratio; as a result, item-level predictors such as base frequency and family size were operationalized as categorical condition factors rather than continuous trial-level covariates. This distinction reflects standard practice in ERP research and aligns the statistical treatment of predictors with the structure of the underlying data.

2.2 Individual Differences and Grouping Variables

To quantify individual differences in linguistic experience and orthographic knowledge, participants completed three standardized measures—the Nelson–Denny Vocabulary Test (Brown et al., 1993), a spelling task adapted from Sawi (2016), and the Author Recognition Test (ART; Stanovich and West 1989). Scores on these measures were moderately correlated (rs = .29–.51, all $p < .02$), indicating partially overlapping but dissociable components of lexical experience.

To capture the shared and distinct variance among these measures, we conducted a principal components analysis (PCA) on standardized (z-scored) vocabulary, spelling, and ART scores. PCA was selected to derive orthogonal dimensions of individual differences without imposing a priori weighting or categorical group boundaries. Two components had eigenvalues greater than 1, jointly accounting for 83.5% of the total variance (58.0% for Dimension 1 and 25.5% for Dimension 2).

Component loadings (Table 1) indicated that Dimension 1 loaded most strongly on vocabulary and print exposure (ART), with a secondary contribution from spelling accuracy, and was interpreted as indexing *Language Proficiency*. Dimension 2 loaded most strongly on spelling accuracy, with weaker and oppositely signed contributions from vocabulary and ART, and was interpreted as indexing *Orthographic Sensitivity*:

- *Dimension 1 (Language Proficiency)*: vocabulary (.82), ART (.81), spelling (.65)
- *Dimension 2 (Orthographic Sensitivity)*: spelling (.76), vocabulary ($-.29$), ART ($-.32$)

The two components were statistically independent ($r \approx 0$), supporting their interpretation as orthogonal dimensions of reading-related skill. These continuous component scores were used as predictors in all behavioral and ERP analyses.

2.3 Statistical Analyses

The analyses are organized around the study's primary theoretical aim, determining whether Orthographic Sensitivity and Language Proficiency differentially modulate the timing and nature of morphological processing. We predicted that Orthographic Sensitivity would primarily modulate early, form-based processing indexed by the N250, while Language Proficiency would primarily modulate later semantic integration indexed by the N400. The behavioral and ERP results are reported separately for words and nonwords, followed by a summary that evaluates these predictions against the obtained pattern.

Table 1. PCA loadings and variance explained for the three lexical-experience measures.

Measure	Dimension 1 (Language Proficiency)	Dimension 2 (Orthographic Sensitivity)
Vocabulary (z_vcb)	0.82	−0.29
Spelling (z_spl)	0.65	0.76
Author Recognition Test (z_art)	0.81	−0.32
Eigenvalue	1.74	0.76
Variance explained (%)	58.0	25.5

2.3.1 Behavioral Data (Reaction Times)

Reaction time (RT) analyses were conducted on correct responses only. RTs were log-transformed to reduce positive skew and to better meet model assumptions of normality and homoscedasticity. Separate analyses were conducted for word and nonword trials: for words, base frequency and family size were the primary morphological predictors; for nonwords, morphological complexity (real vs. pseudo-suffix) was the key manipulation, with base frequency and family size included as covariates (see Aims and Predictions).

RTs were analyzed using linear mixed-effects models fit with the `lme4` package (Bates et al., 2015) in R (R Core Team, 2026). All models included random intercepts for participants and items, and random slopes for within-participant predictors where supported by the data. Continuous lexical predictors (log base frequency and log family size) were mean-centered and scaled prior to analysis. Individual-difference measures—Orthographic Sensitivity and Language Proficiency—were entered as continuous predictors corresponding to standardized PCA scores.

For word trials, models tested the effects of base frequency, family size, Orthographic Sensitivity, and Language Proficiency, as well as interactions between Orthographic Sensitivity and lexical familiarity measures. For nonword trials, models additionally included morphological complexity (simple vs. complex) as a categorical predictor and tested whether complexity effects were modulated by individual differences. Treatment (reference) coding was used for categorical predictors in RT analyses. Model significance was assessed using Type III tests with Satterthwaite-approximated degrees of freedom. Fixed-effect estimates are reported alongside model-based estimated marginal means where relevant. Means are reported in milliseconds following back-transformation from the log scale used in analysis.

2.3.2 ERP Data

For word stimuli, ERP trials were averaged separately according to median splits on base frequency and morphological family size, enabling separate analyses of each variable at the N250 and N400. For nonword stimuli, trials were averaged according to a median split on morphological family size.

The choice of primary item-level variable differed across stimulus types for theoretically motivated reasons. For word stimuli, base frequency was treated as the primary morphological variable of interest in the ERP analyses, because it indexes the strength of the stored lexical representation, that is, how well-entrenched the stem form is in the reader's mental lexicon as a function of accumulated encounter frequency. This makes it the most theoretically relevant item-level variable for examining individual differences in how stored lexical representations are accessed and processed at each ERP stage. For nonword stimuli, morphological family size was treated as the primary item-level variable, because nonwords have no stored whole-word representations and processing must rely on stem-level morphological properties. Family size indexes the richness of the stem's morphological neighbourhood, that is, how many distinct derived words share the stem and reinforce its semantic core, and is therefore the most theoretically relevant variable for examining how morphological family information is recruited when processing a novel form in the absence of whole-word familiarity. This asymmetry between stimulus types reflects the distinct theoretical roles of the word and nonword analyses; words examine the time course of stored lexical representation access, while nonwords examine morphological generalisation to novel forms.

Within the nonword analyses, family size's statistical role further differed across the two time windows: at the N400 it was a primary predictor of theoretical interest, entering into three-way interactions with Morphological Complexity and each individual difference dimension; at the N250 it was retained as a covariate to reduce item-level variance rather than as a predictor of theoretical interest in its own right. Because base frequency and family size are properties of the stem shared across both members of each complexity pair (e.g., *gasful* and *gasfil* share the same stem), neither variable could confound the complexity manipulation. Family size was preferred over base frequency as the N250 covariate on empirical grounds: preliminary model comparisons indicated that family size accounted for substantially more item-level variance in the nonword N250 data (residual variance: 4.07 vs. 27.47). The pattern of interactions reported below was equivalent in direction when base frequency was used as the covariate instead.

ERP analyses were conducted on mean amplitudes computed from condition-averaged waveforms. Separate datasets were constructed for words and nonwords, and analyses were conducted independently for the N250 and N400 time windows. Because ERP amplitudes were derived from condition-level averages rather than trial-level data, morphological predictors in ERP analyses were treated as categorical condition factors rather than continuous covariates.

ERP amplitudes were analysed using linear mixed-effects models. For word stimuli, separate models were fit for base frequency and family size, each entered as the primary morphological predictor along with Orthographic Sensitivity, Language Proficiency, and their interaction with the morphological predictor. The non-interacting individual difference dimension was included as a covariate in each model: Language Proficiency was a covariate in the N250 models, and Orthographic Sensitivity was a covariate in the N400 models. For nonword stimuli, model structure differed across time windows to reflect the distinct role of family size at each stage. At the N250, models included fixed effects of Morphological Complexity, Orthographic Sensitivity, Language Proficiency, interactions between Complexity and each individual difference dimension, and Morphological Family Size as a covariate. At the N400, a single symmetric model was fit including fixed effects of Morphological Complexity, Morphological Family Size, Language Proficiency, Orthographic Sensitivity, and all two- and three-way interactions among Complexity, Family Size, and each individual difference dimension separately. This symmetric structure permitted direct comparison of the roles of LP and OS in modulating the joint influence of morphological complexity and family size on N400 amplitude, providing a formal test of the predicted dissociation between the two dimensions rather than assuming it in the model specification. All models included random intercepts for participants and for electrode site nested within participant, allowing baseline amplitude differences across electrodes to be modelled while preserving the spatial structure of the data.

Categorical predictors in ERP analyses were effect-coded (sum-to-zero coding), such that model intercepts represent grand mean amplitudes and main effects reflect average effects across conditions. Orthographic Sensitivity and Language Proficiency were entered as continuous, mean-centred predictors. Time window was not included as a factor in the models; instead, separate models were fit for the N250 and N400 to preserve component-specific interpretation.

Statistical significance of fixed effects was assessed using Type III F-tests with Satterthwaite-approximated degrees of freedom as implemented in the `lmerTest` package (Kuznetsova et al., 2017). Where higher-order interactions involving individual-difference measures were significant, follow-up analyses were conducted using estimated marginal means (`emmeans`). Interactions involving Orthographic Sensitivity and Language Proficiency were probed by estimating contrasts across the observed range of each dimension in the sample, allowing direct tests of whether morphological effects differed as a function of reader skill and characterising the full extent of any reversals across the individual difference continua.

2.3.3 Model Estimation and Inference

All mixed-effects models were fit using restricted maximum likelihood estimation. To ensure stable convergence, models were estimated using the BOBYQA optimizer. Continuous predictors were centered to facilitate interpretation of interaction terms and model intercepts. For predictors with one degree of freedom, F-tests from the analysis-of-variance tables are mathematically equivalent to squared t-tests from model summaries; all inferential conclusions are therefore invariant to the choice of reporting format. Across all analyses, inferential emphasis was placed on theoretically motivated interactions rather than exhaustive post hoc comparisons. Follow-up contrasts were conducted only where warranted by significant interactions and were limited to tests directly addressing the study's hypotheses. Estimated marginal means and confidence intervals were obtained using the `emmeans` package (Lenth and Piaskowski, 2026); confidence intervals are based on asymptotic approximation, which is appropriate given the sample size.

3 RESULTS

The analyses are organised around two complementary sources of evidence bearing on the study's primary theoretical aim — determining whether Orthographic Sensitivity and Language Proficiency differentially modulate the timing and nature of morphological processing. Reaction time models included individual difference measures as main effects only, reflecting the theoretical framing of behavioural responses as indices of overall processing efficiency. Interactions between morphological variables and individual differences were examined in the ERP models, where the temporal resolution of the N250 and N400 allows these effects to be attributed to specific processing stages. Results are reported separately for words and nonwords. For each stimulus type, behavioural results are presented first, followed by ERP results organised by component, N250 then N400, reflecting the temporal sequence of morphological processing. This organisation reflects the distinct theoretical roles of the two stimulus types — the word analyses examine the time course of base frequency and family size effects in stored lexical representations, while the nonword analyses examine whether morphological sensitivity generalises to novel forms and whether individual differences modulate that generalisation. Within each section, the pattern of effects is evaluated against the predictions derived from morpho-orthographic decomposition accounts and the Lexical Quality Hypothesis, with particular attention to whether Orthographic Sensitivity primarily modulates early form-based processing and Language Proficiency primarily modulates later semantic integration.

Table 2 summarises the fixed effects and interaction terms included in each model. For word stimuli, separate ERP models were run for base frequency and family size, with an asymmetric interaction structure in which Orthographic Sensitivity is the primary moderator at the N250, reflecting early form-based processing, and Language Proficiency is the primary moderator at the N400, reflecting later semantic integration; the non-interacting individual difference dimension was included as a covariate in each model. For nonword stimuli, the N250 model included interactions between morphological complexity and both individual difference dimensions, with family size included as a covariate. The nonword N400 model used a symmetric structure testing both OS and LP within the same analysis, permitting direct comparison of the two dimensions and providing a formal test of the predicted dissociation. Reaction time models included individual difference measures as main effects only, reflecting the theoretical framing of behavioural responses as indices of overall processing efficiency rather than processing architecture.

Table 2. Summary of linear mixed-effects models. OS = Orthographic Sensitivity; LP = Language Proficiency; BF = Base Frequency; FS = Family Size; Cmplx = Morphological Complexity. All models fit by REML using the `bobyqa` optimizer. Degrees of freedom estimated using Satterthwaite's method. RE = random effects; Subj = participant; Elec = electrode site nested within participant. † = included as covariate only. ‡ = reported briefly in main text alongside the corresponding base frequency model; full model output in Supplementary Table S1. § = as ‡; full model output in Supplementary Table S2. ¶ = exploratory model reversing the interaction structure of the corresponding primary model, testing whether the non-predicted individual difference dimension also modulates the effect; full model output in Supplementary Tables S3a–S3b (N250) and S4a–S4b (N400).

Stimulus	**Outcome**	**Variable**	**Fixed effects**	**Interaction(s)**	**RE structure**
Words	RT	BF + FS	BF, FS, OS, LP	None	Subj, Item
Words	N250	BF	BF, OS, LP†	BF × OS	Subj, Elec/Subj
Words	N250	BF×LP¶	BF, LP, OS†	BF × LP	Subj, Elec/Subj
Words	N250	FS‡	FS, OS, LP†	FS × OS	Subj, Elec/Subj
Words	N250	FS×LP¶	FS, LP, OS†	FS × LP	Subj, Elec/Subj
Words	N400	BF	BF, LP, OS†	BF × LP	Subj, Elec/Subj
Words	N400	BF×OS¶	BF, OS, LP†	BF × OS	Subj, Elec/Subj
Words	N400	FS§	FS, LP, OS†	FS × LP	Subj, Elec/Subj
Words	N400	FS×OS¶	FS, OS, LP†	FS × OS	Subj, Elec/Subj
Nonwords	RT	BF + FS	Cmplx, BF, FS, OS, LP	None	Subj (Cmplx slope), Item
Nonwords	N250	FS†	Cmplx, OS, LP, FS†	Cmplx × OS, Cmplx × LP	Subj, Elec/Subj
Nonwords	N400	FS	Cmplx, FS, LP, OS	Cmplx × FS × LP, Cmplx × FS × OS	Subj, Elec/Subj

The model structure reflected in Table 2 warrants brief explanation. For word stimuli, separate ERP models were run for base frequency and family size at each time window, reflecting the fact that these variables were operationalised as separate median-split factors in distinct datasets. At the N250, both the base frequency and family size models tested the interaction with Orthographic Sensitivity, consistent with the prediction that early form-based processing is primarily modulated by sublexical orthographic knowledge; Language Proficiency was included as a covariate in both models. The corresponding N400 models reversed this structure, testing interactions with Language Proficiency and including Orthographic Sensitivity as a covariate. This asymmetric structure for word stimuli reflects the theoretical predictions developed in the Aims and Predictions section: OS is predicted to modulate early form-based processing indexed by the N250, while LP is predicted to modulate later semantic integration indexed by the N400.

To assess whether the non-predicted individual difference dimension nonetheless modulated processing at each component, exploratory models were also fit reversing the interaction structure at each time window. At the N250, Base Frequency × LP and Family Size × LP were each tested with Orthographic Sensitivity included as a covariate. At the N400, Base Frequency × OS and Family Size × OS were each tested with Language Proficiency included as a covariate. These exploratory analyses parallel the logic of the symmetric nonword N400 model described below, testing rather than assuming the absence of cross-dimension modulation at each component.

For nonword stimuli, base frequency was included as a covariate in the RT model but was not independently examined in the ERP analyses. At the N250, morphological family size was included as a covariate to reduce item-level variance rather than as a predictor of theoretical interest; the primary fixed effects were Morphological Complexity and its interactions with both individual difference dimensions, reflecting the prediction that early processing of novel forms is modulated by reader skill. At the N400, family size was elevated to a primary predictor, entering into three-way interactions with Morphological Complexity and each individual difference dimension simultaneously. Both OS and LP were included as interacting variables within the same symmetric model, permitting direct comparison of their roles in modulating the joint influence of morphological complexity and family size on N400 amplitude, and providing a formal test of the predicted dissociation between the two dimensions. This distinction in the role of family size across the two nonword ERP models reflects the theoretical rationale developed in the Aims and Predictions section: family size indexes the richness of the morphological neighbourhood and is predicted to modulate semantic integration specifically at the N400, where the precision of lexical-semantic representations — indexed by LP — determines whether that richness facilitates or burdens processing. The symmetric model structure ensures that this prediction is tested against the alternative that OS, rather than LP, drives the family size modulation, rather than assuming the dissociation in the model specification itself.

3.1 Words

Words RT

Reaction times (RTs) for correct lexical decisions to word stimuli were analyzed using a linear mixed-effects model with fixed effects of Base Frequency, Family Size, Orthographic Sensitivity (OS), and Language Proficiency (LP). The model included random intercepts for participants and items. The analysis revealed robust main effects of item-level lexical familiarity. As base frequency increased, recognition times decreased, $\beta = -0.017$, $SE = 0.005$, $t(100.97) = -3.66$, $p < .001$, $\eta_p^2 = .12$. Similarly, words from larger morphological families elicited faster responses than those from smaller families, $\beta = -0.013$, $SE = 0.005$, $t(93.59) = -2.64$, $p = .010$, $\eta_p^2 = .07$. Orthographic Sensitivity showed a marginal trend toward faster overall response times, $\beta = -0.035$, $SE = 0.019$, $t(63.84) = -1.85$, $p = .069$, $\eta_p^2 = .05$. Language Proficiency did not exert a reliable main effect on RTs, $\beta = 0.012$, $SE = 0.012$, $t(63.97) = 0.94$, $p = .35$. The model explained 3.3% of the variance via fixed effects ($R^2_{\text{marginal}} = .033$) and 41.7% overall ($R^2_{\text{conditional}} = .417$). These findings indicate that word recognition speed is driven primarily by item-level lexical familiarity; Orthographic Sensitivity shows only a marginal and unstable influence on overall response efficiency and does not selectively modulate morphological effects on response times. (See Figure 1)

Words N250: Base Frequency

Mean N250 amplitudes for word stimuli were analyzed using a linear mixed-effects model with fixed effects of Base Frequency (high vs. low), Orthographic Sensitivity (OS), Language Proficiency (LP), and the interaction between Base Frequency and OS. The model included random intercepts for participants

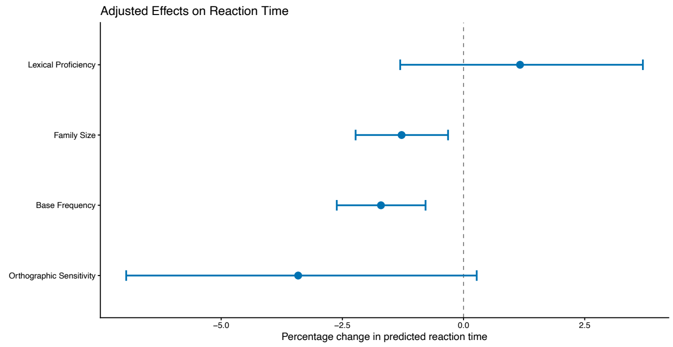

Figure 1. Estimated regression coefficients and 95% confidence intervals for subject the variables Lexical Proficiency, Orthographic Sensitivity,and the item variables Morphological Family Size and Base Frequeny. Point estimates (dots) show the effect size for each predictor variable, with horizontal whiskers representing 95% confidence intervals. The vertical dashed line marks the null effect at zero. Variables with intervals not crossing zero are statistically significant.

and for electrode site nested within participant. The analysis revealed a significant main effect of Base Frequency—words from higher-frequency bases elicited more negative N250 amplitudes than words from lower-frequency bases, $M_{\text{high}} = -1.63\,\mu$V, 95% CI $[-2.56, -0.69]$, $M_{\text{low}} = -1.29\,\mu$V, 95% CI $[-2.22, -0.36]$, $\beta = -0.318$, $SE = 0.100$, $t(1618) = -3.18$, $p = .002$, $\eta_p^2 = .006$. Neither Orthographic Sensitivity, $\beta = 0.534$, $SE = 0.520$, $t(57) = 1.03$, $p = .309$, nor Language Proficiency, $\beta = -0.536$, $SE = 0.378$, $t(57) = -1.42$, $p = .162$, yielded reliable main effects. Critically, the interaction between Base Frequency and Orthographic Sensitivity was significant, $\beta = -0.455$, $SE = 0.112$, $t(1618) = -4.06$, $p < .001$, $\eta_p^2 = .010$, indicating that the magnitude of early base frequency effects is modulated by orthographic skill. The negative direction of the interaction indicates that readers higher in Orthographic Sensitivity showed an amplified N250 base frequency effect—a larger difference in early processing amplitude between high- and low-frequency stems—relative to readers lower in OS, consistent with the interpretation that more precise sublexical orthographic representations strengthen early morpheme-level activation signatures. The model explained 1.8% of variance via fixed effects ($R^2_{\text{marginal}} = .018$) and 80.1% overall ($R^2_{\text{conditional}} = .801$). (See Figure 2 A)

A parallel analysis conducted on family size showed a similar pattern: words from smaller morphological families elicited more negative N250 amplitudes than words from larger families, $\beta = 0.324$, $SE = 0.058$, $t(1618) = 5.54$, $p < .001$, $\eta_p^2 = .02$, consistent with the form-meaning distinction developed in the Discussion, whereby richer family structure provides top-down semantic support that reduces early processing demands. The Family Size $\times$ OS interaction showed a marginal trend in the same direction as the base frequency interaction above, $\beta = -0.111$, $SE = 0.065$, $t(1618) = -1.70$, $p = .089$, but did not reach significance. (See Figure 2 B) Full model output for the family size analysis is reported in Supplementary Table S1.

To assess whether Language Proficiency also modulated early form-based processing, an exploratory model reversing the interaction structure was fit for the N250 word data, with Orthographic Sensitivity included as a covariate. The interaction between Base Frequency and Language Proficiency was significant, $\beta = 0.180$, $SE = 0.082$, $t(1618) = 2.20$, $p = .028$. Follow-up contrasts across the observed LP range indicated that this modulation ran in the opposite direction to the Orthographic Sensitivity interaction reported above: rather than amplifying with reader skill, the base frequency effect was largest at low LP (LP $= -2.5$ SD: $\Delta = -0.78\,\mu$V, $t(1618) = -3.47$, $p = .001$) and shrank steadily as LP increased, remaining significant at mean LP ($\Delta = -0.33\,\mu$V, $t(1618) = -3.27$, $p = .001$) but becoming non-significant by

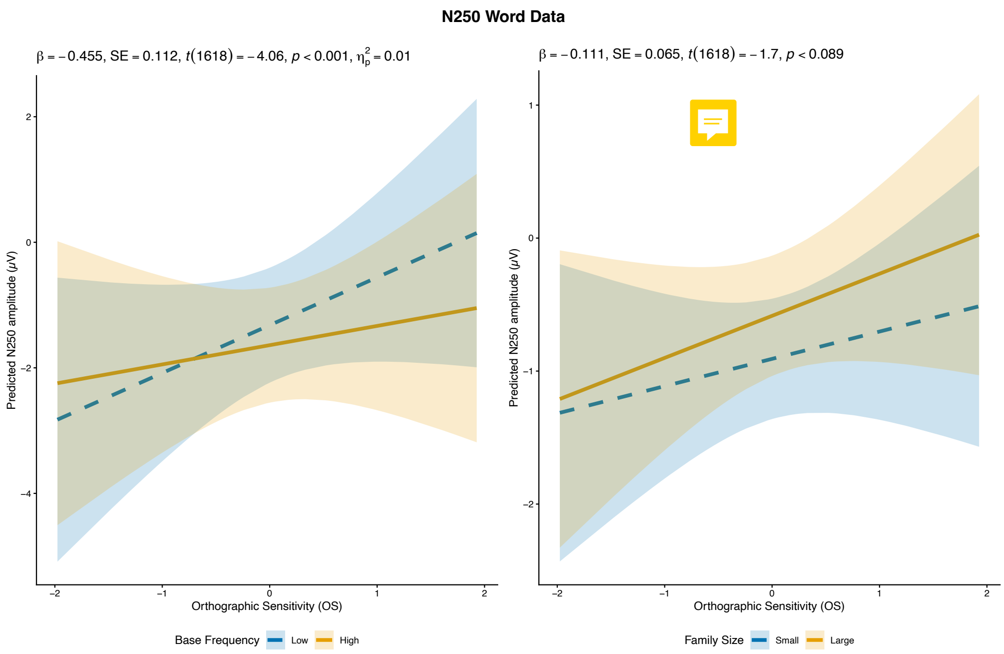

Figure 2. Predicted N250 amplitude as a function of Orthographic Sensitivity (OS), plotted separately for base frequency (a) and morphological family size (b) in the words analyses. (a) The Base Frequency × OS interaction: the base frequency effect is minimal at low OS but becomes pronounced at high OS, with high-frequency bases eliciting more negative amplitudes than low-frequency bases. (b) The corresponding Family Size × OS interaction shows a similar but attenuated pattern that does not reach significance. Shaded ribbons represent 95% confidence intervals; points reflect model-predicted values.

LP $= +1$ SD ($\Delta = -0.15\,\mu$V, $t(1618) = -1.13$, $p = .259$), with the point estimate reversed but not significant at the upper end of the observed range (LP $= +3$ SD: $\Delta = 0.21\,\mu$V, $t(1618) = 0.78$, $p = .433$). A parallel exploratory model testing the interaction between Family Size and Language Proficiency at the N250 found no significant modulation, $\beta = -0.074$, $SE = 0.047$, $t(1618) = -1.56$, $p = .119$; full output for both exploratory N250 models is reported in Supplementary Tables S3a–S3b.

Words N400: Base Frequency

Mean N400 amplitudes for word stimuli were analysed using a linear mixed-effects model with fixed effects of Base Frequency (high vs. low), Language Proficiency (LP), Orthographic Sensitivity (OS), and the interaction between Base Frequency and LP. The model included random intercepts for participants and for electrode site nested within participant. The analysis revealed a significant main effect of Base Frequency — averaged across LP levels, words from higher-frequency bases elicited more positive N400 amplitudes than words from lower-frequency bases, $M_{\text{high}} = 0.71\,\mu$V, 95% CI $[-0.44, 1.86]$, $M_{\text{low}} = 0.52\,\mu$V, 95% CI $[-0.63, 1.66]$, $\beta = 0.214$, $SE = 0.106$, $t(1618) = 2.01$, $p = .044$, $\eta_p^2 = .003$. A significant main effect of Orthographic Sensitivity was also observed, $\beta = 1.290$, $SE = 0.640$, $t(57) = 2.02$, $p = .049$, $\eta_p^2 = .07$, reflecting overall differences in N400 amplitude as a function of Orthographic Sensitivity, independently of morphological structure — a pattern consistent with the OS main effect observed in the family size model below. Although this effect was significant, the p-value was close to the conventional threshold ($p = .049$), and should be interpreted with appropriate caution. Language Proficiency did not yield a reliable main effect, $\beta = 0.047$, $SE = 0.465$, $t(57) = 0.10$, $p = .920$.

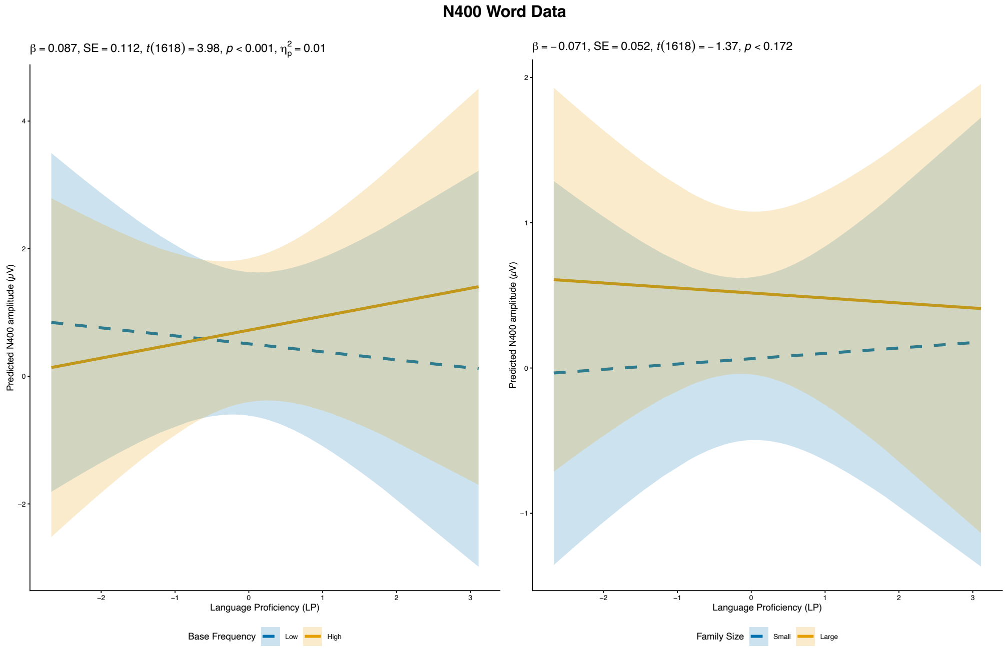

Figure 3. Predicted N400 amplitude as a function of Language Proficiency (LP), plotted separately for base frequency (a) and morphological family size (b) in the words analyses. (a) The Base Frequency × LP interaction: the base frequency effect is minimal at low LP but grows substantially at high LP, with high-frequency bases eliciting more positive amplitudes than low-frequency bases (b) The corresponding Family Size × LP interaction is not significant. Shaded ribbons represent 95% confidence intervals; points reflect model-predicted values

Critically, the interaction between Base Frequency and Language Proficiency was significant, $\beta = 0.344$, $SE = 0.087$, $t(1618) = 3.98$, $p < .001$, $\eta_p^2 = .010$. Follow-up contrasts examined the base frequency effect at LP values from -2 to $+2$ SD, spanning most of the observed LP range (-2.67 to $+3.11$). The direction of the base frequency effect reversed across the LP continuum. At LP $= -2$ SD, high base frequency words elicited more negative N400 amplitudes than low base frequency words ($\Delta = -0.470\,\mu$V, $t(1618) = -2.38$, $p = .018$; EMM: High $= 0.28\,\mu$V, Low $= 0.76\,\mu$V). At LP $= -1$ SD the contrast was not significant ($\Delta = -0.130\,\mu$V, $t(1618) = -0.97$, $p = .333$). At mean LP the direction had reversed, with high base frequency words now eliciting more positive N400 amplitudes than low base frequency words ($\Delta = 0.210\,\mu$V, $t(1618) = 2.01$, $p = .044$; EMM: High $= 0.72\,\mu$V, Low $= 0.51\,\mu$V), and this pattern strengthened monotonically at LP $= +1$ SD ($\Delta = 0.560\,\mu$V, $t(1618) = 3.99$, $p < .001$; EMM: High $= 0.94\,\mu$V, Low $= 0.38\,\mu$V) and LP $= +2$ SD ($\Delta = 0.900\,\mu$V, $t(1618) = 4.37$, $p < .001$; EMM: High $= 1.16\,\mu$V, Low $= 0.26\,\mu$V). The participant LP scores ranged from -2.67 to $+3.11$, confirming that estimates at ± 2 SD are grounded in observed data rather than extrapolation. Together, these contrasts indicate that LP modulates the contribution of base frequency to N400 amplitude in a manner that crosses over across the LP continuum, with the nature of that modulation shifting from a pattern in which high base frequency words are harder to process semantically at low LP, to one in which they are easier to process at high LP. The model explained 2.6% of variance via fixed effects ($R^2_{\text{marginal}} = .026$) and 82.5% overall ($R^2_{\text{conditional}} = .825$). (See Figure 3 A)

A parallel analysis conducted on family size revealed a robust main effect, with words from smaller morphological families eliciting greater N400 amplitudes than words from larger families, $\beta = 0.452$, $SE = 0.064$, $t(1618) = 7.07$, $p < .001$, $\eta_p^2 = .03$, consistent with facilitated semantic integration for words supported by richer morphological families. A significant main effect of Orthographic Sensitivity was also observed in this model, $\beta = 0.677$, $SE = 0.318$, $t(57) = 2.13$, $p = .038$, $\eta_p^2 = .07$, mirroring the OS main effect reported above. Unlike base frequency, however, family size did not interact with Language Proficiency, $\beta = -0.071$, $SE = 0.052$, $t(1618) = -1.37$, $p = .172$, suggesting that LP does not selectively modulate the semantic contribution of family size for familiar words, plausibly because stored whole-word representations reduce the need to recruit family-level semantic information during access. (See Figure 3 B) Full model output for the family size analysis is reported in Supplementary Table S2.

To assess whether Orthographic Sensitivity also modulated semantic integration, parallel exploratory models reversing the interaction structure were fit for the N400 word data, with Language Proficiency included as a covariate. Neither the interaction between Base Frequency and Orthographic Sensitivity, $\beta = 0.195$, $SE = 0.120$, $t(1618) = 1.63$, $p = .104$, nor the interaction between Family Size and Orthographic Sensitivity, $\beta = 0.091$, $SE = 0.072$, $t(1618) = 1.27$, $p = .204$, was significant, indicating that, unlike Language Proficiency, Orthographic Sensitivity did not modulate either morphological variable at the N400 for word stimuli; full model output is reported in Supplementary Tables S4a–S4b.

3.2 Nonwords

Nonwords RT

Reaction times (RTs) for correct lexical decisions to nonwords were analyzed using a linear mixed-effects model with fixed effects of Morphological Complexity (simple vs. complex), Base Frequency, Family Size, Orthographic Sensitivity (OS), and Language Proficiency (LP). The model included random intercepts for participants and items and by-participant random slopes for Complexity. The analysis revealed a strong main effect of Morphological Complexity—complex nonwords elicited slower responses than simple nonwords, $M_{\text{complex}} = 718.9$ ms, 95% CI $[695.2, 743.4]$, $M_{\text{simple}} = 684.6$ ms, 95% CI $[661.5, 708.5]$, $\beta = 0.049$, $SE = 0.005$, $t(62.68) = 9.13$, $p < .001$, $\eta_p^2 = .57$. (See Figure 4 A) In addition, Base Frequency exerted a reliable inhibitory effect, $\beta = 0.015$, $SE = 0.005$, $t(98.21) = 3.05$, $p = .003$, $\eta_p^2 = .09$, such that nonwords derived from higher-frequency bases were responded to more slowly than those from lower-frequency bases. Family Size did not significantly influence response times, $\beta = -0.006$, $SE = 0.005$, $t(96.64) = -1.24$, $p = .22$. Orthographic Sensitivity showed a significant main effect, with higher-sensitivity participants responding more quickly overall, $\beta = -0.041$, $SE = 0.018$, $t(62.21) = -2.23$, $p = .030$, $\eta_p^2 = .07$. In contrast, there was no effect of Language Proficiency on nonword RTs, $\beta = 0.003$, $SE = 0.012$, $t(62.91) = 0.28$, $p = .78$. The model explained 5.0% of the variance via fixed effects ($R^2_{\text{marginal}} = .050$) and 49.4% overall ($R^2_{\text{conditional}} = .494$). Taken together, these results indicate that nonword decision latencies are strongly shaped by morphological complexity and item-level lexical familiarity, particularly base frequency, while individual differences primarily affect overall response efficiency, with more orthographically sensitive readers responding more quickly across conditions. (See Figure 4)

Nonwords N250

Mean N250 amplitudes for nonword stimuli were analysed using a linear mixed-effects model with fixed effects of Morphological Complexity, Orthographic Sensitivity (OS), Language Proficiency (LP), interactions between Complexity and each individual difference dimension, and Morphological Family Size as a covariate. The model included random intercepts for participants and for electrode site nested within participant.

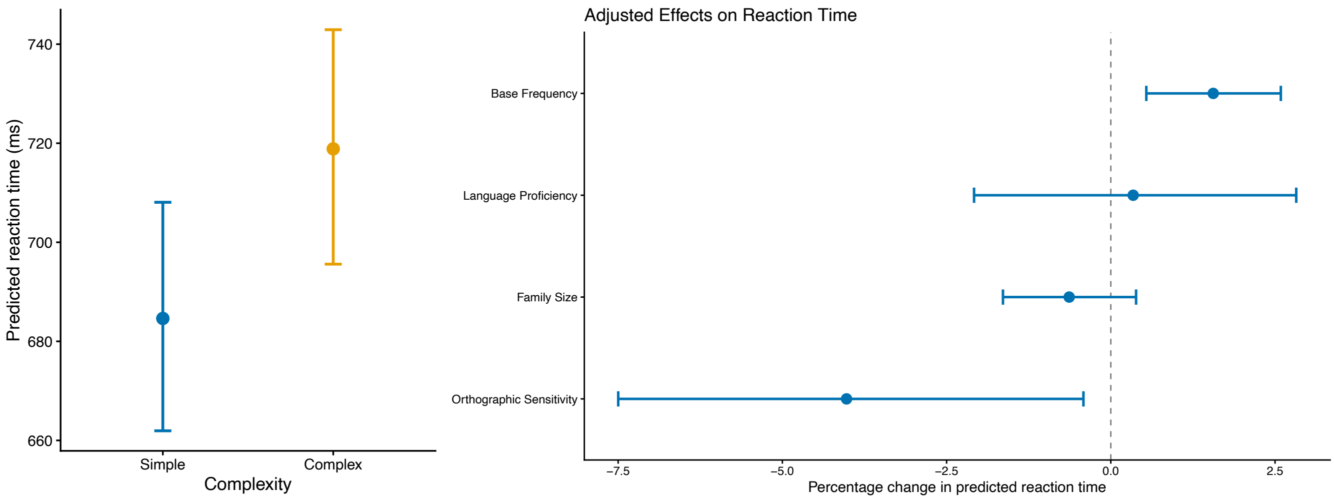

Figure 4. Nonword reaction time results. (a) Predicted reaction time (ms) by complexity condition (Simple vs Complex), showing the significant complexity cost. (b) Adjusted percentage change in predicted reaction time associated with each predictor (Base Frequency, Family Size, Language Proficiency, Orthographic Sensitivity), with error bars representing confidence intervals. Complexity and Base Frequency were associated with significant increases in reaction time, and Orthographic Sensitivity with a significant decrease; Family Size and Language Proficiency were not significant predictors.

The analysis revealed a significant main effect of Morphological Complexity, with complex nonwords eliciting more negative N250 amplitudes than simple nonwords overall ($M_{\text{complex}} = -0.759\,\mu$V $[-1.156, -0.363]$, $M_{\text{simple}} = -0.526\,\mu$V $[-0.922, -0.130]$), $\beta = -0.233$, $SE = 0.050$, $t(4856) = -4.63$, $p < .001$. Morphological Family Size, included as a covariate, yielded a significant effect, $\beta = 0.182$, $SE = 0.050$, $t(4856) = 3.63$, $p < .001$, with large-family nonwords eliciting less negative amplitudes than small-family nonwords. Neither OS nor LP yielded reliable main effects, $\beta = 0.298$, $SE = 0.220$, $t(57) = 1.35$, $p = .181$; $\beta = -0.207$, $SE = 0.160$, $t(57) = -1.29$, $p = .202$, respectively.

The Complexity $\times$ OS interaction was significant, $\beta = 0.392$, $SE = 0.056$, $t(4856) = 6.98$, $p < .001$, $\eta_p^2 = .010$. Follow-up contrasts examined the complexity effect across the observed OS range (-1.9 to $+1.9$ SD). The complexity effect was large and reliable at OS $= -1.9$ SD ($\Delta = 0.990\,\mu$V, $t = 8.29$, $p < .001$) and at OS $= -1$ SD ($\Delta = 0.642\,\mu$V, $t = 8.33$, $p < .001$), attenuated but still significant at mean OS ($\Delta = 0.250\,\mu$V, $t = 4.98$, $p < .001$), and reversed in direction but only marginally significant at OS $= +1$ SD ($\Delta = -0.142\,\mu$V, $t = -1.93$, $p = .053$). At the upper extreme of the observed OS range (OS $= +1.9$ SD) the reversal was significant ($\Delta = -0.490\,\mu$V, $t = -4.27$, $p < .001$). Examination of estimated marginal means indicated that this reversal was driven primarily by the trajectory of complex nonwords, which decreased in negativity across the OS range (EMM: $-1.720\,\mu$V at OS $= -1.9$ SD to $+0.160\,\mu$V at OS $= +1.9$ SD), while simple nonwords also decreased in negativity but more slowly (EMM: $-0.720\,\mu$V at OS $= -1.9$ SD to $-0.340\,\mu$V at OS $= +1.9$ SD). (See Figure 5 A)

The Complexity $\times$ LP interaction was also significant, $\beta = 0.358$, $SE = 0.041$, $t(4856) = 8.76$, $p < .001$, $\eta_p^2 = .016$. Follow-up contrasts examined the complexity effect across the observed LP range (-2.5 to $+3.0$ SD). The complexity effect was large and reliable at LP $= -2.5$ SD ($\Delta = 1.110\,\mu$V, $t = 9.92$, $p < .001$) and at LP $= -1$ SD ($\Delta = 0.574\,\mu$V, $t = 9.05$, $p < .001$), attenuated but still significant at mean LP ($\Delta = 0.216\,\mu$V, $t = 4.31$, $p < .001$), and significantly reversed at LP $= +1$ SD ($\Delta = -0.141\,\mu$V, $t = -2.14$, $p = .032$) and strongly reversed at LP $= +3$ SD ($\Delta = -0.860\,\mu$V, $t = -6.38$, $p < .001$). In contrast to the OS pattern, examination of estimated marginal means indicated

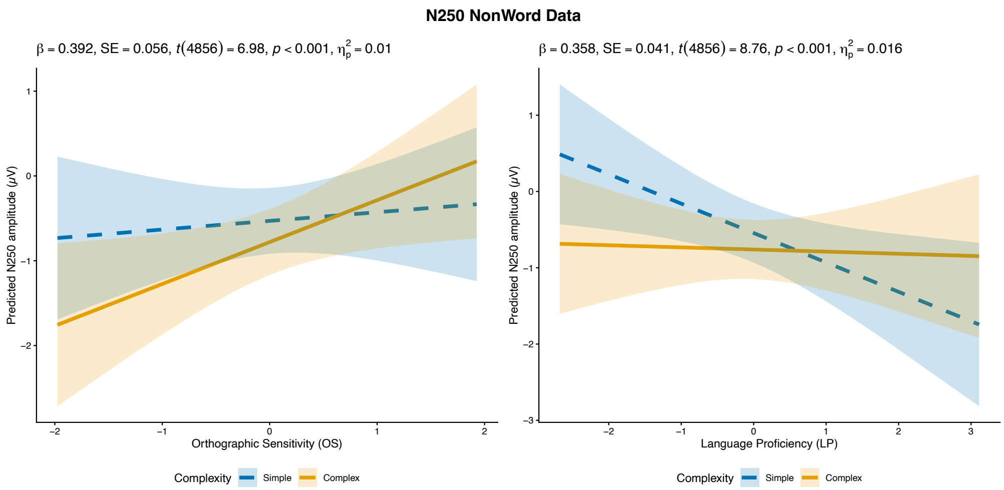

Figure 5. Predicted N250 amplitude for nonwords as a function of Orthographic Sensitivity (OS; a) and Language Proficiency (LP; b), plotted separately by complexity (Simple vs Complex). (a) The Complexity × OS interaction: the complexity cost (Complex more negative than Simple) is large at low OS, attenuates with increasing OS, and reverses at high OS. (b) The parallel Complexity × LP interaction shows the same pattern: the complexity cost is large at low LP and reverses at high LP. Shaded ribbons represent 95% confidence intervals; points reflect model-predicted values.

that the LP reversal was driven primarily by the trajectory of simple nonwords, which became increasingly negative across the LP range (EMM: $+0.420\,\mu$V at LP $= -2.5$ SD to $-1.700\,\mu$V at LP $= +3.0$ SD), while complex nonwords remained essentially flat (EMM: $-0.690\,\mu$V at LP $= -2.5$ SD to $-0.840\,\mu$V at LP $= +3.0$ SD). Participant LP scores ranged from -2.67 to $+3.11$ and OS scores ranged from -1.98 to $+1.93$, confirming that all contrasts are grounded in observed data rather than extrapolation. Model fit indices were $R^2_{\text{marginal}} = .025$ and $R^2_{\text{conditional}} = .575$. (See Figure 5 B)

Nonwords N400

Mean N400 amplitudes for nonword stimuli were analysed using a single symmetric linear mixed-effects model including fixed effects of Morphological Complexity, Morphological Family Size, Language Proficiency (LP), Orthographic Sensitivity (OS), and all interactions among Complexity, Family Size, and each individual difference dimension separately. The model included random intercepts for participants and for electrode site nested within participant. This symmetric structure permitted direct comparison of the roles of LP and OS in modulating the joint influence of morphological complexity and family size at the N400, providing a formal test of the predicted dissociation between the two dimensions.

The analysis revealed a significant main effect of Morphological Complexity, with complex nonwords eliciting more negative N400 amplitudes than simple nonwords, $\beta = -0.469$, $SE = 0.061$, $t(4851) = -7.64$, $p < .001$, $\eta^2_p = .010$. Family Size did not yield a reliable main effect, $\beta = 0.085$, $SE = 0.061$, $t(4851) = 1.38$, $p = .168$. Neither LP, $\beta = 0.011$, $SE = 0.255$, $t(57) = 0.045$, $p = .964$, nor OS, $\beta = 0.381$, $SE = 0.350$, $t(57) = 1.09$, $p = .281$, yielded reliable main effects.

The Complexity $\times$ Family Size interaction was significant, $\beta = -0.446$, $SE = 0.123$, $t(4851) = -3.63$, $p < .001$, indicating that the N400 complexity effect was larger for nonwords from large morphological families than for those from small families, averaged across both individual difference dimensions. The Complexity $\times$ LP interaction was significant, $\beta = 0.254$, $SE = 0.050$, $t(4851) = 5.08$, $p < .001$, $\eta^2_p = .005$. Follow-up contrasts examined the complexity effect across the observed LP range (-2.5 to

$+3.0$ SD), averaged over family size. The complexity effect was large and reliable at LP $= -2.5$ SD ($\Delta = 1.090\,\mu$V, $t = 7.99$, $p < .001$) and at LP $= -1$ SD ($\Delta = 0.710\,\mu$V, $t = 9.19$, $p < .001$), attenuated at mean LP ($\Delta = 0.460\,\mu$V, $t = 7.48$, $p < .001$), and further reduced but still significant at LP $= +1$ SD ($\Delta = 0.210\,\mu$V, $t = 2.54$, $p = .011$). At the upper limit of the observed LP range (LP $= +3$ SD) the complexity effect reversed in direction but did not reach significance ($\Delta = -0.300\,\mu$V, $t = -1.84$, $p = .065$). The Family Size $\times$ LP interaction was also significant, $\beta = 0.122$, $SE = 0.050$, $t(4851) = 2.45$, $p = .015$, indicating that the influence of family size on N400 amplitude varied as a function of LP; this interaction is qualified by the three-way interaction reported below. The Complexity $\times$ OS interaction was significant, $\beta = 0.240$, $SE = 0.069$, $t(4851) = 3.49$, $p < .001$, $\eta_p^2 = .003$, indicating that the complexity effect also attenuated with increasing OS, though as reported below, OS did not selectively modulate the joint influence of complexity and family size. The Family Size $\times$ OS interaction was not significant, $\beta = 0.066$, $SE = 0.069$, $t(4851) = 0.955$, $p = .340$.

Critically, the three-way Complexity $\times$ Family Size $\times$ LP interaction was significant, $\beta = 0.229$, $SE = 0.100$, $t(4851) = 2.29$, $p = .022$, $\eta_p^2 = .001$, while the corresponding Complexity $\times$ Family Size $\times$ OS interaction was not significant, $\beta = 0.143$, $SE = 0.138$, $t(4851) = 1.04$, $p = .300$. This asymmetry — LP driving the three-way interaction while OS does not — constitutes the primary evidence for the predicted dissociation between the two individual difference dimensions at the N400.

Follow-up interaction contrasts for the significant LP three-way examined the Complexity $\times$ Family Size interaction separately across the observed LP range (-2.5 to $+3.0$ SD). At LP $= -2.5$ SD, the Complexity $\times$ Family Size interaction was large and highly reliable ($\Delta = -1.010\,\mu$V, $t = -3.70$, $p < .001$) — the complexity effect was substantially larger for large-family nonwords ($\Delta = 1.600\,\mu$V, $t = 8.26$, $p < .001$) than for small-family nonwords ($\Delta = 0.590\,\mu$V, $t = 3.03$, $p = .002$), reflecting disproportionately strong semantic processing demands when both morphological complexity and family richness are high. At LP $= -1$ SD the interaction remained large and significant ($\Delta = -0.670\,\mu$V, $t = -4.31$, $p < .001$), with complexity effects again larger for large-family ($\Delta = 1.050\,\mu$V, $t = 9.55$, $p < .001$) than small-family nonwords ($\Delta = 0.380\,\mu$V, $t = 3.45$, $p = .001$). At mean LP the interaction remained significant but attenuated ($\Delta = -0.440\,\mu$V, $t = -3.58$, $p < .001$), with complexity effects larger for large-family ($\Delta = 0.680\,\mu$V, $t = 7.82$, $p < .001$) than small-family nonwords ($\Delta = 0.240\,\mu$V, $t = 2.76$, $p = .006$). At LP $= +1$ SD the interaction was no longer significant ($\Delta = -0.210\,\mu$V, $t = -1.31$, $p = .192$), and at LP $= +3$ SD — the upper limit of the observed LP range — the interaction reversed in sign but did not reach significance ($\Delta = 0.250\,\mu$V, $t = 0.76$, $p = .450$).

To characterise the nature of the LP modulation more precisely, follow-up contrasts examined the family size effect within complex nonwords specifically across the observed LP range. At LP $= -2.5$ SD, large-family complex nonwords elicited substantially more negative N400 amplitudes than small-family complex nonwords ($\Delta = 0.720\,\mu$V, $t = 3.74$, $p < .001$). This difference decreased monotonically with increasing LP: at LP $= -1$ SD ($\Delta = 0.370\,\mu$V, $t = 3.36$, $p = .001$), at mean LP ($\Delta = 0.130\,\mu$V, $t = 1.53$, $p = .127$), and at LP $= +1$ SD the difference was negligible and non-significant ($\Delta = -0.100\,\mu$V, $t = -0.91$, $p = .360$). At LP $= +3$ SD the pattern reversed significantly — small-family complex nonwords now elicited more negative N400 amplitudes than large-family complex nonwords ($\Delta = -0.580\,\mu$V, $t = -2.49$, $p = .013$) — indicating that for highly proficient readers, morphological family richness facilitates rather than amplifies semantic integration demands for complex nonwords. (See Figure 6)

Although the Complexity $\times$ Family Size $\times$ OS interaction was not significant, $\beta = 0.143$, $SE = 0.138$, $t(4851) = 1.04$, $p = .30$, descriptive contrasts were conducted across the observed OS range (-1.9 to $+1.9$ SD) to characterise the nature of this null result and to directly compare the OS pattern with the LP

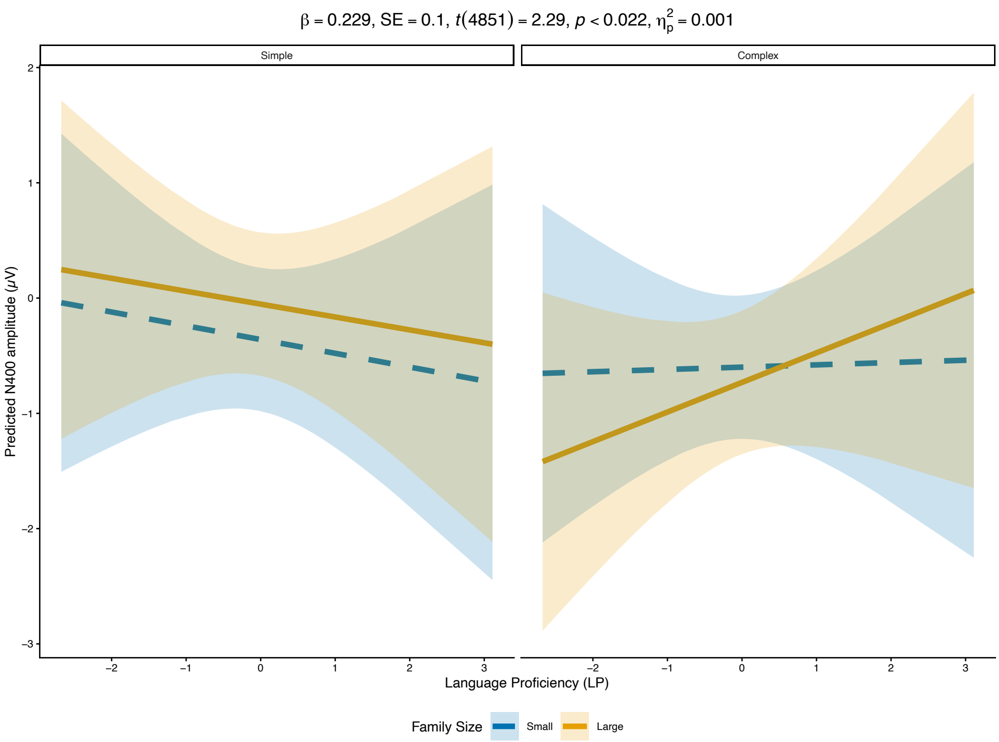

Figure 6. Predicted N400 amplitude for nonwords as a function of Language Proficiency (LP), plotted separately by family size (Small vs Large) within each complexity condition (Simple, Complex), illustrating the Complexity × Family Size × LP three-way interaction. Within simple nonwords, the family size effect is small and relatively stable across the LP range. Within complex nonwords, the pattern shifts systematically with LP: large-family nonwords elicit substantially more negative amplitudes than small-family nonwords at low LP, this difference diminishes through the mid-range of LP, and at the upper end of the observed range the pattern reverses, with small-family nonwords eliciting more negative amplitudes than large-family nonwords. Shaded ribbons represent 95% confidence intervals.

pattern reported above. The Complexity $\times$ Family Size interaction was significant at low OS (OS $= -1.9$ SD: $\Delta = -0.730\,\mu$V, $t = -2.48$, $p = .013$; OS $= -1$ SD: $\Delta = -0.600\,\mu$V, $t = -3.18$, $p = .001$) and at mean OS ($\Delta = -0.460\,\mu$V, $t = -3.72$, $p < .001$), reflecting a pattern qualitatively similar to that seen at low LP. However, unlike the LP pattern, the interaction attenuated monotonically without reversing as OS increased: at OS $= +1$ SD the interaction was marginal ($\Delta = -0.310\,\mu$V, $t = -1.75$, $p = .081$), and at the upper extreme of the observed OS range (OS $= +1.9$ SD) it was absent ($\Delta = -0.190\,\mu$V, $t = -0.66$, $p = .512$). Crucially, the family size effect within complex nonwords showed no significant reversal at any level of OS — at OS $= +1.9$ SD the contrast was negligible and non-significant ($\Delta = -0.110\,\mu$V, $t = -0.55$, $p = .581$). This pattern of attenuation without reversal for OS, in contrast to the significant reversal at high LP, constitutes the clearest expression of the dissociation between the two dimensions at the N400. Participant LP scores ranged from -2.67 to $+3.11$ and OS scores ranged from -1.98 to $+1.93$, confirming that all contrasts are grounded in observed data rather than extrapolation. Model fit indices were $R^2_{\text{marginal}} = .013$ and $R^2_{\text{conditional}} = .667$. (See Figure 7)

4 DISCUSSION

This study examined whether individual differences in Orthographic Sensitivity (OS) and Language Proficiency (LP) differentially modulate morphological processing at distinct temporal stages, as indexed

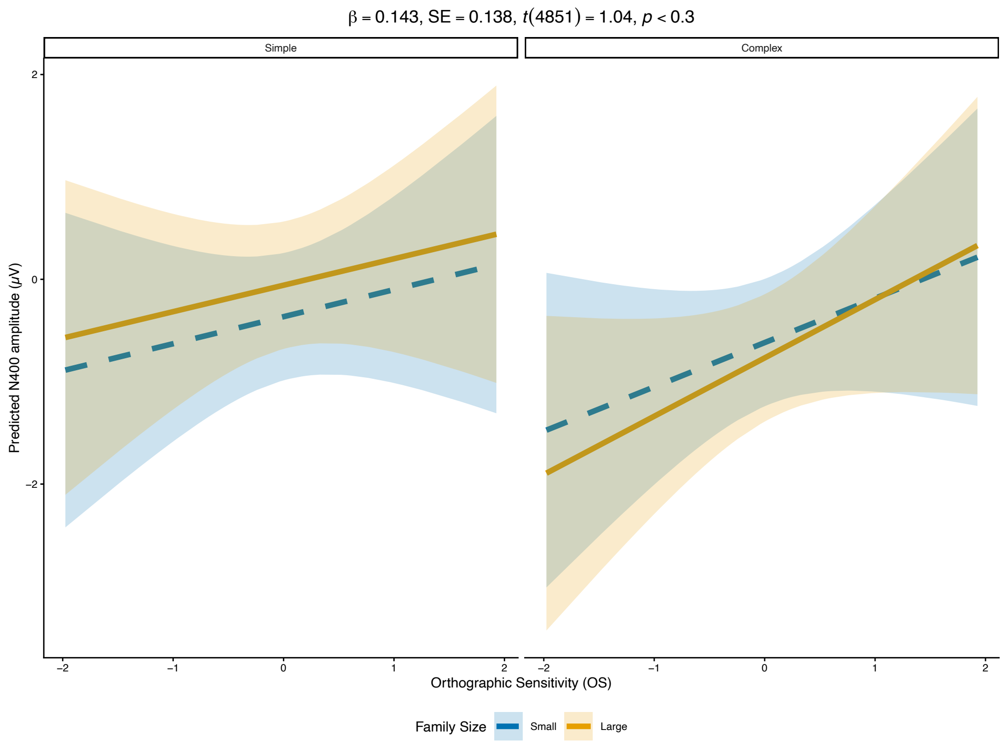

Figure 7. Predicted N400 amplitude for nonwords as a function of Orthographic Sensitivity (OS), plotted separately by family size (Small vs Large) within each complexity condition (Simple, Complex). Unlike the corresponding LP three-way interaction (Figure 2c), the Complexity × Family Size × OS interaction was not significant, indicating that OS does not selectively modulate the joint influence of morphological complexity and family size at the N400. This null result supports the specificity of the LP effect and constitutes part of the evidence for the predicted dissociation between the two individual difference dimensions. Shaded ribbons represent 95% confidence intervals.

by the N250 and N400 components of the ERP. Three findings structure the discussion that follows. First, individual differences were largely absent from behavioural response times, with the exception of an OS advantage for nonword lexical decisions, indicating that the two dimensions differ in their contribution to general lexical decision efficiency, but that neither dimension selectively influenced morphological effects in reaction times. Second, at the N250, both OS and LP modulated the complexity effect for nonword stimuli through distinct mechanisms that are developed in detail below, and OS primarily modulated the base frequency effect for word stimuli, with a smaller LP effect also present, running in the opposite direction, providing partial support for the predicted role of orthographic skill in early morpho-orthographic parsing. Third, at the N400, a model testing both OS and LP within the same analysis revealed that LP selectively drove the three-way interaction between morphological complexity and family size for nonword stimuli, while the corresponding OS interaction was not significant, constituting the clearest evidence for the predicted dissociation between the two dimensions. LP also shaped base frequency effects for word stimuli: the facilitatory effect of high base frequency, a more positive N400 relative to low frequency, was absent for readers low in LP but grew larger as LP increased, consistent with LP determining how form-level frequency information is weighted during semantic integration. Together these findings support a partial dissociation between OS and LP in morphological processing, with a mechanistically rich picture at the N250 and the clearest dissociation emerging at the N400.

4.1 Early Morpho-orthographic Processing: The N250

4.1.1 Word Stimuli

For word stimuli, base frequency modulated the N250 as a function of Orthographic Sensitivity: high base frequency words elicited more negative amplitudes than low base frequency words, and this difference increased with OS, a pattern consistent with the prediction that early form-based processing is selectively shaped by the precision of sublexical orthographic representations. Morphological family size also modulated the N250 as a main effect, with large-family words eliciting less negative amplitudes than small-family words. A Family Size $\times$ OS interaction showed the same qualitative trend as base frequency, though it fell short of significance ($p = .089$), suggesting that OS may extend to family size sensitivity as well, but this possibility remains tentative pending further evidence.

The opposite directions of the base frequency and family size effects at the N250 are theoretically informative and can be accounted for by a distinction between form-level entrenchment and semantic reinforcement of morphological structure. Base frequency reflects how often an orthographic form has been encountered as the same lexical entry, across inflectional variants that share a stable meaning but do not create new form-meaning pairings at the lexical level. Family size, by contrast, reflects how many distinct derived words embed the base, each constituting a separate lexical entry that pairs the base form with a related but distinct meaning. Repeated encounter of the same form across multiple such meaning-bearing contexts is argued to build richer, more precise semantic representations (Bolger et al., 2008; Li et al., 2026) as the semantic core of the base becomes more robustly specified as it is reinforced across independent form-meaning pairings. Under this account, it is the quality of the lexical representation associated with large family size, that is, the precision and richness of the semantic core built through accumulated contextual variation, that enables stronger top-down semantic feedback to the form level, facilitating orthographic identification and reducing N250 amplitude. Base frequency does not carry this semantic reinforcement; repeated encounters with a high-frequency base accumulate orthographic familiarity without necessarily building new form-meaning pairings, and its effect at the N250 therefore reflects engagement of morpho-orthographic parsing that is not necessarily supplemented by feedback from a richly specified semantic representation.

The modulation of this base frequency effect by OS is consistent with the theoretical profile of this dimension. OS indexes sensitivity to sublexical orthographic structure, and readers higher in OS are more sensitive to the statistical and structural properties of sub-word units, including stem forms. For these readers, a high-frequency base strongly recruits morpho-orthographic parsing activity, producing greater N250 amplitude. The absence of a corresponding modulation by LP is consistent with the prediction that early form-based processing is specifically tied to orthographic skill rather than lexical-semantic knowledge, and supports the theoretical distinction between the two individual difference dimensions at this processing stage. As discussed below, however, the nonword data reveal a more complex picture at the N250, with LP also modulating early morphological parsing through a mechanistically distinct route.

4.1.2 Nonword Stimuli

For nonword stimuli, both OS and LP modulated the N250 complexity effect, though via independent routes rather than a single shared mechanism operating at different strengths. This constitutes a partial qualification of the predicted OS-specific modulation at the N250, but the nature of the qualification is itself theoretically informative.

For readers lower in OS, complex nonwords elicited substantially more negative N250 amplitudes than simple nonwords, reflecting a large early processing cost associated with the presence of a real suffix. As OS increased, this complexity cost attenuated monotonically, reversing significantly at the upper extreme of the observed OS range (OS $= +1.9$ SD: $\Delta = -0.490\,\mu$V, $t = -4.27$, $p < .001$), though not at OS $= +1$ SD ($\Delta = -0.142\,\mu$V, $t = -1.93$, $p = .053$). Examination of estimated marginal means indicated that this reversal was driven primarily by the trajectory of complex nonwords, which decreased in negativity across the OS range (EMM: $-1.720\,\mu$V at OS $= -1.9$ SD to $+0.160\,\mu$V at OS $= +1.9$ SD), while simple nonwords also decreased in negativity but more slowly (EMM: $-0.720\,\mu$V at OS $= -1.9$ SD to $-0.340\,\mu$V at OS $= +1.9$ SD). We propose that this pattern reflects affix-led parsing (Taft and Forster, 1975; Rastle and Davis, 2008), in which the presence of a real suffix triggers morpho-orthographic parsing activity across all readers, but the efficiency with which that parsing proceeds varies with OS. On this account, for readers lower in OS, the parse is triggered by the real suffix but does not proceed efficiently, producing a processing cost for complex nonwords relative to simple ones; as OS increases, parsing becomes more efficient, reducing and eventually reversing the complexity cost. At very high OS, on this view, the parse is efficient enough that complex nonwords become easier to process than simple ones, which lack a real suffix and so do not benefit from affix-based parsing in the same way. If this account is correct, the reversal reflects the complex nonword trajectory outpacing the simple nonword trajectory as OS increases, rather than a rising processing cost for simple nonwords.

The LP pattern showed a different statistical signature, and the contrast with the OS pattern points to two distinct sources of the reversal within the nonword N250 data. As with OS, the complexity effect reversed at high LP — significantly so at LP $= +1$ SD ($\Delta = -0.141\,\mu$V, $t = -2.14$, $p = .032$) and more strongly at LP $= +3$ SD ($\Delta = -0.860\,\mu$V, $t = -6.38$, $p < .001$). However, examination of estimated marginal means indicated that a different trajectory underlies this reversal than the one underlying the OS reversal. Complex nonwords changed little across the LP range (EMM: $-0.690\,\mu$V at LP $= -2.5$ SD to $-0.820\,\mu$V at LP $= +3.0$ SD), suggesting that LP had minimal effect on the processing of complex nonwords. Instead, the reversal was driven primarily by simple nonwords becoming increasingly negative as LP increased, from $+0.420\,\mu$V at LP $= -2.5$ SD to $-1.700\,\mu$V at LP $= +3.0$ SD. We propose that this pattern reflects stem-led parsing: readers higher in LP have richer and more precise representations of stems as free morphemes with discrete lexical entries, and recognition of the stem in a nonword triggers an attempt to assign morphological structure to the whole string regardless of what follows. This proposal is consistent with Beyersmann et al. (2016), who found that high proficiency readers showed embedded stem priming effects independent of whether the stem was followed by a real affix or a non-affixal ending, a pattern suggesting that stem activation in skilled readers is not contingent on affix recognition, in contrast to the standard affix-stripping account (Taft and Forster, 1975; Rastle and Davis, 2008). On this account, for complex nonwords the stem-led parse succeeds—a real stem is followed by a real suffix, yielding a valid morphological analysis—and successful parsing produces neither a cost nor a benefit at the N250, keeping amplitude stable across LP levels. For simple nonwords, on this account, the parse fails—the pseudoaffix cannot be integrated into a valid morphological structure—and this failed parse produces a processing cost that grows with LP, driving the simple nonword trajectory increasingly negative.

The contrast between the two reversal patterns is informative. The OS reversal appears to be driven by the complex nonword trajectory: the real suffix triggers morpho-orthographic parsing in all readers, but higher OS readers resolve that parse more efficiently, driving complex nonwords to decrease in negativity faster than simple nonwords and eventually producing a reversal; simple nonwords would not be expected to show this pattern, since they do not contain a real suffix for affix-based parsing efficiency to act on. The LP reversal, by contrast, appears to be driven by the simple nonword trajectory: stem recognition may trigger an attempt to assign morphological structure to the whole string in higher LP readers, an attempt that succeeds for complex nonwords, keeping their amplitude stable across LP levels, but fails for simple nonwords, producing a processing cost that grows with LP. This pattern is broadly consistent with the theoretical profiles of the two dimensions: OS indexes the efficiency of sublexical orthographic processing, and its effect on early parsing is expressed specifically in the complex nonword condition, where the presence of a real affix drives differential parsing efficiency across OS levels; LP indexes the depth and precision of lexical-semantic knowledge, which we suggest may extend to knowledge of stems as free morphemes with discrete lexical entries, such that its effect on early parsing is expressed specifically in the simple nonword condition, where stem recognition triggers a parse that does not succeed, producing a cost that grows with LP. The concurrent modulation of the N250 complexity effect by both OS and LP is consistent with Beyersmann et al. (2016)'s suggestion that affix-chunking and embedded stem activation mechanisms may operate concurrently during early morphological segmentation, with the present data extending this proposal by suggesting that these two mechanisms are dissociable across individual difference dimensions: OS appears to index the efficiency of affix-based parsing, while LP appears to index the propensity for stem-based parsing independent of affix status.

This mechanistic distinction between OS and LP at the N250 for nonwords is further grounded by the word data reported above. For familiar words, OS modulated the base frequency effect at the N250 — a finding that reflects OS operating at the level of stem-form entrenchment in stored lexical representations. The specificity of the affix-led account for nonwords therefore does not imply that OS is exclusively sensitive to affixes in all contexts; rather, it suggests that when readers encounter a novel form for which no stored whole-word representation is available, OS-driven parsing activity is channelled specifically through affix recognition, because it is the affix that provides the primary morphological signal in that context. LP, by contrast, operates through stem recognition in both familiar and novel form processing, consistent with its profile as an index of the depth and precision of lexical-semantic representations in which stems are the primary meaning-bearing units.

4.2 Semantic Integration and the N400

The N400 findings provide the clearest support for the predicted dissociation between Orthographic Sensitivity and Language Proficiency. LP selectively modulated N400 amplitude in interaction with morphological variables across both word and nonword stimuli, consistent with the theoretical prediction that depth of lexical-semantic knowledge determines how morphological information is used during meaning access. OS modulated the overall N400 complexity effect for nonwords, reflecting a general reduction in semantic processing demands as orthographic processing becomes more efficient, but did not modulate the joint influence of morphological complexity and family size on N400 amplitude. To understand why the three-way Complexity $\times$ Family Size $\times$ LP interaction — and not the corresponding OS interaction — constitutes the primary test of the predicted dissociation at the N400, it is necessary to be clear about what family size indexes and why LP is predicted to modulate its influence specifically at this later processing stage. When a reader encounters a morphologically complex form, the stem may activate semantic information associated with other words in its morphological family — derived words that share the stem and preserve its core meaning. The N400 reflects how easily that semantic activation is accessed and integrated. LP, as an index of the depth and precision of lexical-semantic representations, is predicted to determine whether the semantic richness of the morphological neighbourhood facilitates or burdens that integration process. OS, as an index of sublexical orthographic processing efficiency, is not predicted to selectively modulate this semantic process, and the data are consistent with that prediction. The asymmetry between LP and OS at the N400, established within a single model testing both dimensions simultaneously, provides the formal basis for the dissociation claim developed in the sections that follow.

4.2.1 Word Stimuli

For word stimuli, the direction of the base frequency effect at the N400 depended on LP: readers lower in LP showed greater N400 negativity for high than low frequency bases, while readers higher in LP showed the opposite pattern, with high frequency bases associated with less negative N400 amplitudes than low frequency bases. This pattern was not significant at LP $= -1$ SD but was significant at the lower and upper ends of the tested LP range, with the effect of base frequency in opposite directions at LP $= -2$ SD and LP $= +2$ SD.

This finding is consistent with the Lexical Quality Hypothesis (Perfetti, 2007), under which more proficient readers have more precise and fully specified lexical representations — involving tighter connections among orthographic, phonological, and semantic information — while less proficient readers have less precise representations in which these connections are less fully specified. Under this account, LP modulates how base frequency contributes to N400 amplitude during semantic access: for readers lower in LP, higher base frequency is associated with greater N400 negativity, suggesting that base frequency influences how actively meaning retrieval is pursued; for readers higher in LP, higher base frequency is associated with less N400 negativity, suggesting that a more familiar orthographic form supports easier meaning retrieval. The direction of the base frequency effect therefore shifts across the LP continuum, with the pattern at low LP and the pattern at high LP reflecting different expressions of how form-level familiarity interacts with the precision of lexical representations during semantic access.

We note that the present data do not allow us to determine precisely why the direction of the base frequency effect shifts across LP levels. One possibility is that low and high LP readers differ in the degree to which they attempt meaning retrieval for words of varying familiarity — low LP readers mounting a more active retrieval attempt for familiar high frequency forms and a less thorough one for less familiar low frequency forms, while high LP readers retrieve meaning efficiently regardless of base frequency but do so more easily for more familiar forms. Another possibility is that the shift reflects a single continuous process in which LP modulates the weight given to base frequency during semantic access, with the direction of the effect changing as a mathematical consequence of that modulation across the LP range. The linear interaction term in the model is consistent with both accounts and cannot distinguish between them. Accuracy data would help to adjudicate: if low LP readers are less likely to correctly recognise low frequency words — treating them as nonwords more often — that would support the first account; if accuracy is comparable across base frequency levels for both LP groups, the second account would be more plausible.

4.2.2 Nonword Stimuli

The three-way Complexity $\times$ Family Size $\times$ LP interaction illustrates the mechanistic dissociation between OS and LP at the N400, and is most readily understood by first considering why morphological family size might influence N400 amplitude at all. When a reader encounters a complex nonword like *gasful*, the stem *gas* activates other words in its morphological family — *gases*, *gassy*, *gaseous* — each of which shares the stem's core meaning. A stem with a large morphological family activates more semantic associates than one with a small family, creating a richer semantic network that must be evaluated against a form that has no stored lexical entry. Whether this richer activation constitutes a processing burden or a processing advantage depends on two factors simultaneously, as the three-way interaction reveals: whether the form signals a valid morphological structure — which determines whether the stem's morphological family is recruited during processing — and the precision of the reader's lexical-semantic representations, indexed by LP. For simple nonwords, the pseudoaffix does not signal a valid morphological structure. Without that morphological signal, the reader does not parse the string as a derived form and does not recruit the stem's morphological family as part of semantic processing. If family-level semantic activation is not recruited, its richness cannot influence processing, and LP — which determines how efficiently that activation is leveraged — has nothing to modulate. This is why the family size effect for simple nonwords is small and does not vary with LP. For complex nonwords, by contrast, the real suffix signals valid morphological structure, triggering recruitment of the stem's morphological family and making LP the critical determinant of whether that activation facilitates or burdens processing.

Figure 6 illustrates how LP modulates the family size effect specifically for complex nonwords. The figure plots N400 amplitude as a function of LP separately for large and small family nonwords, with simple and complex nonwords shown in separate panels. For simple nonwords, the family size lines are relatively flat and parallel across the LP range, consistent with the account above. For complex nonwords, the picture is strikingly different: as LP increases, the amplitude for large-family complex nonwords decreases substantially, while the amplitude for small-family complex nonwords remains relatively stable. At low LP, large-family complex nonwords elicit considerably more negative N400 amplitudes than small-family complex nonwords; at high LP, this difference reverses, with large-family complex nonwords eliciting less negative amplitudes than small-family ones. It is this divergence of the large-family and small-family trajectories specifically within complex nonwords, and the stability of those trajectories within simple nonwords, that constitutes the three-way interaction.

This pattern is interpretable within the Lexical Quality Hypothesis (Perfetti, 2007), under which more proficient readers have more precise and fully specified lexical representations, involving tighter connections among orthographic, phonological, and semantic information. Under this framework, what differs across LP levels is not the quantity of semantic activation generated by a large morphological family, but the quality of the representations that are activated. High LP readers have precise, well-specified representations of the stem's morphological relatives — each family member is a distinct, fully specified lexical entry with a clear form-meaning relationship. When such a reader encounters a large-family complex nonword, the activated semantic network is organised and coherent, and the reader can draw on this organised knowledge to efficiently evaluate the novel form: they recognise the morphological structure, identify the stem and its semantic core, and process the novel derived form against a well-specified representational background. The result is less negative N400 amplitude for large-family complex nonwords at high LP — the richness of the family facilitates rather than burdens processing. For readers lower in LP, the same network is activated but the representations of family members are less precise and less fully specified. The reader cannot draw on this activation as efficiently, and the additional semantic content generated by a large family creates processing demands that are difficult to resolve against a novel form with no lexical entry. The result is more negative N400 amplitude for large-family complex nonwords at low LP — the richness of the family amplifies rather than reduces processing demands.

This account is also consistent with the distinction between base frequency and family size developed in the context of the N250 findings. Family size reflects the richness of the semantic core built through repeated encounter of a stable form with a stable meaning across multiple distinct form-meaning pairings (Bolger et al., 2008; Li et al., 2026). It is precisely this semantic richness that makes family size a double-edged variable at the N400: for readers whose representations are insufficiently precise to leverage it, semantic richness amplifies processing demands; for readers whose representations are sufficiently precise, it provides a processing advantage. Base frequency, which accumulates orthographic familiarity without building new form-meaning pairings, does not carry this semantic richness, and its role at the N400 is correspondingly different — modulating how form-level familiarity contributes to semantic access rather than determining whether morphological family richness facilitates or burdens processing.

The OS pattern at the N400 stands in informative contrast to the LP pattern. OS reduced the overall complexity effect at the N400 — higher OS readers showed a smaller difference in N400 amplitude between complex and simple nonwords, consistent with more efficient orthographic processing reducing general semantic processing demands. However, the three-way Complexity × Family Size × OS interaction was not significant, indicating that OS did not modulate the family size effect for complex nonwords in the way that LP did. Unlike the LP pattern, where the family size effect within complex nonwords reversed significantly at high LP, the corresponding OS pattern showed only attenuation — the family size difference for complex nonwords decreased as OS increased but did not reverse at any point across the observed OS range. This dissociation — LP driving a reversal of the family size effect within complex nonwords while OS produces only attenuation — is the clearest expression of the predicted dissociation between the two dimensions at the N400. It suggests that while orthographic processing efficiency influences the general demands of semantic processing, it is the precision of lexical-semantic representations — indexed by LP — that specifically determines whether morphological family richness facilitates or burdens semantic integration of novel morphologically complex forms. Together, the word and nonword N400 findings converge on a coherent picture: LP indexes the precision of lexical-semantic representations, and it is this precision that determines how morphological information — whether the orthographic entrenchment of a familiar base or the semantic richness of a large morphological family — influences semantic access during morphological processing.

4.3 Behavioral and Neural Measures of Individual Differences in Morphological Processing

A consistent pattern across the present results is that individual differences in OS and LP were largely absent from behavioral response times but systematically present in the ERP components. This dissociation is not a null result but a theoretically meaningful finding: RT and ERPs index different aspects of lexical processing. Response times reflect the speed and outcome of the decision process — how quickly a reader reaches a response — while ERP components reflect the dynamics of processing as it unfolds in time, making visible the stages and mechanisms that contribute to that decision without being reducible to its speed.

For OS, the confirmatory RT models showed that higher OS predicted faster overall responding for nonword stimuli, consistent with orthographic processing efficiency contributing to the speed of lexical decisions for novel forms. However, OS did not selectively modulate how morphological structure influenced response times — the complexity effect and the base frequency effect were equivalent across OS levels in the confirmatory models. This pattern suggests that OS affects how efficiently orthographic information is processed without changing how morphological structure influences the decision process at the behavioral level. The ERP data tell a different story: OS modulated the base frequency effect at the N250 for words, and drove the affix-led reversal of the complexity effect at the N250 for nonwords. These effects are real and theoretically interpretable, but they do not propagate to the behavioral outcome in the form of differential morphological effects on RT.

For LP, the confirmatory RT models showed no reliable effect on overall response times for either words or nonwords. LP did not predict overall lexical decision speed, nor did it modulate morphological effects in the confirmatory models. Again, the ERP data reveal a different picture: LP drove the stem-led reversal of the complexity effect at the N250, and selectively modulated the three-way Complexity × Family Size interaction at the N400. These are systematic and theoretically interpretable effects that are simply not visible in RT.

An exploratory model adding Complexity × OS and Complexity × LP interactions to the nonword RT model clarifies this picture further. The Complexity × OS interaction was not significant in the exploratory RT model ($\beta = -0.002$, $SE = 0.006$, $t(59) = -0.358$, $p = .721$), confirming that the OS modulation of morphological complexity visible at the N250 does not produce differential behavioral responses to complex and simple nonwords. The Complexity × LP interaction was significant ($\beta = 0.009$, $SE = 0.004$, $t(64) = 2.108$, $p = .039$), but in the opposite direction to the N250 pattern. Follow-up contrasts across the observed LP range showed that the complexity cost in RT — complex nonwords responded to more slowly than simple ones — grew monotonically with LP, from $\Delta = 0.026$ log ms at LP $= -2.5$ SD ($t = 2.17$, $p = .034$) to $\Delta = 0.074$ log ms at LP $= +3$ SD ($t = 5.70$, $p < .001$), with no reversal at any LP level. This contrasts sharply with the N250, where the complexity cost decreased and reversed with LP.

Rather than contradicting the neural account, this dissociation between the RT and N250 LP patterns is consistent with the stem-led parsing interpretation developed above. High LP readers engage in more thorough morphological analysis of complex nonwords before reaching a decision — recognising the stem, attempting to assign morphological structure, and evaluating the form against known morphological patterns. This analysis takes more time, producing a larger RT complexity cost. At the neural level, however, the parsing process itself is more efficient for high LP readers — the N250 reversal reflects that successful parsing of complex nonwords becomes facilitative while failed parsing of simple nonwords incurs a cost. The neural efficiency and the behavioral cost are therefore two sides of the same coin: high LP readers process morphological structure more efficiently at the parsing stage, but precisely because they engage more thoroughly with that structure, they take longer to reach a decision for complex nonwords that must ultimately be rejected.

This pattern has methodological implications for how individual differences in morphological processing are measured. Response times alone would suggest that OS and LP have limited and non-specific effects on morphological processing — OS predicts overall nonword decision speed, and LP predicts nothing. The ERP data reveal that both dimensions systematically modulate morphological processing at distinct temporal stages and through distinct mechanisms, in ways that are invisible to RT. This is consistent with the Lexical Quality Hypothesis (Perfetti, 2007), under which the precision of lexical representations affects the quality and architecture of processing rather than simply its speed. Individual differences in representational precision are therefore more readily detected by measures that track processing dynamics — such as ERP components — than by measures that index only the speed and outcome of the decision.

4.4 Theoretical Implications and Limitations

The present findings have several theoretical implications for accounts of morphological processing and individual differences in reading. We consider each in turn before addressing the limitations of the present study.

4.4.1 The Form-Meaning Distinction in Morphological Processing

A recurring theme across the N250 and N400 findings is the theoretical distinction between base frequency and morphological family size as indexing form-level entrenchment and semantic richness respectively. Base frequency reflects how often an orthographic form has been encountered as the same lexical entry, accumulating orthographic familiarity without necessarily building new form-meaning pairings. Family size reflects how many distinct derived words embed the base, each reinforcing the semantic core of the base form across multiple independent form-meaning pairings (Bolger et al., 2008; Li et al., 2026). This distinction accounts for the opposite directions of the two variables at the N250 — base frequency increasing N250 amplitude while family size reduces it — and for their different roles at the N400, where family size but not base frequency enters into the predicted three-way interaction with morphological complexity and LP.

The form-meaning distinction also provides a principled account of why OS influences early processing while LP influences later processing. OS indexes sensitivity to sublexical orthographic structure — the dimension of individual variation most directly relevant to form-level processing — and primarily modulates base frequency effects at the N250, a modulation to which LP contributes only a smaller, oppositely signed effect (reported above; Supplementary Table S3a). LP indexes depth and breadth of lexical-semantic knowledge — the dimension most directly relevant to meaning-level processing — and selectively drives the three-way interaction involving family size at the N400, where OS plays no corresponding role. This asymmetry — OS cleanly restricted to the N250, LP dominant but not fully confined to the N400 — means the two dimensions do not mirror one another symmetrically; rather, each aligns most strongly, though not exclusively, with the variable and processing stage it was predicted to govern.

4.4.2 Affix-Led and Stem-Led Parsing as Dissociable Mechanisms

The nonword N250 findings extend existing accounts of morphological decomposition by showing that affix-based and stem-based parsing mechanisms are dissociable across individual difference dimensions. Affix-stripping accounts (Taft and Forster, 1975; Rastle and Davis, 2008) propose that the affix is identified first and triggers decomposition, after which the stem is looked up in the lexicon. The OS pattern at the N250 is consistent with this account: OS modulates the efficiency with which affix-triggered parsing proceeds, with higher OS readers resolving the parse more efficiently for complex nonwords and showing no corresponding effect for simple nonwords where no real affix is present.

The LP pattern requires a different account. Beyersmann et al. (2016) showed that readers higher in overall reading proficiency demonstrated embedded stem priming independent of affix status, suggesting that stem activation in skilled readers is not contingent on affix recognition. The present findings extend this observation by showing that it is specifically LP — depth and precision of lexical-semantic knowledge — rather than OS that drives stem-based parsing in the N250 data. High LP readers show a growing processing cost for simple nonwords across the LP range, driven primarily by the simple nonword trajectory while complex nonwords remain flat. This pattern is consistent with stem recognition triggering a morphological analysis attempt regardless of what follows, with the parse failing for simple nonwords because the pseudoaffix cannot be integrated into a valid morphological structure. The dissociation between OS and LP at the N250 therefore aligns with the distinction between affix-led and stem-led parsing: OS governs the efficiency of affix-triggered decomposition, while LP governs the propensity for stem-triggered morphological analysis independent of affix status.

This finding has implications for how morpho-orthographic decomposition accounts are framed. Current accounts treat early decomposition as a relatively uniform process driven primarily by affix recognition (Rastle and Davis, 2008), with individual differences not typically incorporated into the theoretical framework. The present results suggest that early parsing is not uniform across readers — it reflects at least two distinct mechanisms that vary independently across individuals, one tied to orthographic sensitivity to affix structure and one tied to the depth of lexical-semantic knowledge in which stems are embedded.

4.4.3 Lexical Quality and the Role of LP at the N400

The N400 findings connect naturally to the Lexical Quality Hypothesis (Perfetti, 2007), under which more proficient readers have more precise and fully specified lexical representations involving tighter connections among orthographic, phonological, and semantic information. The three-way Complexity $\times$ Family Size $\times$ LP interaction for nonwords is a particularly clear expression of this principle: the precision of lexical-semantic representations — indexed by LP — determines whether the semantic richness of a morphological family constitutes a processing burden or a processing advantage during novel form recognition. For low LP readers, a rich morphological neighbourhood creates semantic activation that cannot be efficiently organised against a novel form with no lexical entry, amplifying processing demands. For high LP readers, the same neighbourhood provides well-organised semantic knowledge that facilitates evaluation of the novel form, reducing processing demands. Representational precision therefore determines not just how fast semantic access proceeds but whether morphological richness helps or hinders it.

The word N400 finding — a crossover interaction between base frequency and LP — is also interpretable within this framework, though the mechanism is less precisely specified. LP modulates how base frequency contributes to N400 amplitude in a direction that changes across the LP range, consistent with LP determining how form-level familiarity interacts with the precision of lexical representations during semantic access. The nature of this modulation cannot be fully characterised from the present data alone, and accuracy data would be needed to adjudicate between possible accounts.

4.4.4 Limitations and Future Directions

Several limitations of the present study warrant acknowledgement.

First, the analyses were not pre-registered. The overall analytic strategy was theory-driven and specified in advance — including the symmetric model structure testing both OS and LP at each time window, and the primary predictions about OS at the N250 and LP at the N400. However, several secondary analytic decisions were made in response to patterns in the data, including the extension of follow-up contrasts to the observed limits of the LP and OS distributions, the choice of family size over base frequency as the nonword N250 covariate on empirical grounds, and the exploratory RT model examining Complexity $\times$ LP and Complexity $\times$ OS interactions. These decisions are documented and justified throughout, but the results should nonetheless be treated as requiring replication. Future studies should use the present findings to pre-specify the key analytic decisions before data collection, particularly the predicted three-way Complexity $\times$ Family Size $\times$ LP interaction and the symmetric model structure at the N400.

Second, the word stimuli were not matched to the nonword stimuli on morphological variables, and the two stimulus sets are doing qualitatively different theoretical work — words index the time course of stored lexical representations while nonwords index generalisation to novel forms. Direct comparison of word and nonword findings therefore requires caution, and the theoretical claims developed here rest primarily on the nonword data.

Third, the Base Frequency $\times$ LP crossover at the N400 for words remains mechanistically underspecified. The present data are consistent with LP modulating how form-level familiarity contributes to semantic access, but the nature of that modulation — whether it reflects a shift in the depth of semantic engagement or a continuous change in how base frequency is weighted during access — cannot be determined from N400 amplitude alone. Accuracy data would be needed to characterise the mechanism more precisely, and future studies should collect both RT and accuracy measures to allow fuller interpretation of crossover interactions of this kind.

Fourth, the nonword stimuli were drawn from Sawi (2016) and may not generalise to other nonword sets. The affix-led versus stem-led parsing account developed here rests on the complex versus simple nonword contrast, which holds stem identity constant while manipulating suffix legitimacy. Future studies should replicate this contrast with independently constructed stimulus sets and should consider manipulating stem and affix properties orthogonally to provide more direct tests of the proposed mechanisms.

Fifth, the present sample consisted entirely of adult readers. The developmental trajectory of the OS/LP dissociation — when affix-led and stem-led parsing mechanisms become dissociable, and how this relates to the acquisition of morphological knowledge — remains an open question. Developmental studies examining the emergence of these individual difference effects across the school years would be a valuable extension of the present findings.

Finally, several key effects in the present study have small effect sizes, particularly the three-way Complexity $\times$ Family Size $\times$ LP interaction at the N400 ($\eta_p^2 = .001$). While the effect is theoretically interpretable and statistically reliable, its small magnitude underscores the need for replication and suggests that future studies should be powered specifically to detect three-way interactions of this size. The present results provide the effect size estimates needed for such power calculations.

CONFLICT OF INTEREST STATEMENT

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

AUTHOR CONTRIBUTIONS

JM: Conceptualization, Methodology, Software, Validation, Formal analysis, Resources, Data curation, Visualization, Writing – original draft, Writing – review & editing, Supervision. EK: Methodology, Software, Investigation, Data curation, Writing – original draft, Writing – review & editing, Supervision. FG: Methodology, Software, Investigation, Data curation. JP: Investigation, Data curation, Writing – review & editing TR: Investigation, Data curation, Writing – review & editing EM: Investigation, Data curation, Writing – review & editing CR-M: Investigation, Data curation, Writing – review & editing NA: Investigation, Data curation, Writing – review & editing

All authors contributed to the article and approved the submitted version.

FUNDING

Research reported in this manuscript was supported by the Rhode Island IDeA Network of Biomedical Research Excellence, funded by the National Institute of General Medical Sciences of the National Institutes of Health under grant number P20GM103430. Additional support for student researchers (Joemari Pulido, Tess Rooney, Natalia Alzate, Ethan Moore, and Crismar Ramos-Marte) and for data collection was provided through the Center for Engaged Learning at Providence College and a series of Hampshire College Faculty Development Grants awarded to Dr. Morris during her tenure at that institution. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

ACKNOWLEDGMENTS

We thank Sophie Fulghum, Dylan Palumbo, Maeve Garrity, Aleksandra Kurylowicz, Sarah Debian, Aleksandra Saneff, Kenny Diaz, and Kiley Vallee for their assistance with participant recruitment and data collection.

SUPPLEMENTAL DATA

Supplementary Material should be uploaded separately on submission, if there are Supplementary Figures, please include the caption in the same file as the figure. LaTeX Supplementary Material templates can be found in the Frontiers LaTeX folder.

DATA AVAILABILITY STATEMENT

The datasets [GENERATED/ANALYZED] for this study can be found in the [NAME OF REPOSITORY] [LINK].

REFERENCES

Andrews, S. (2012). Individual differences in skilled visual word recognition and reading: The role of lexical quality. In *Visual Word Recognition: Meaning and Context, Individuals and Development, Vol. 2*, ed. J. Adelman (New York, NY, US: Psychology Press), Current Issues in the Psychology of Language. 151–172

Andrews, S. and Lo, S. (2013). Is morphological priming stronger for transparent than opaque words? It depends on individual differences in spelling and vocabulary. *Journal of Memory and Language* 68, 279–296. doi:10.1016/j.jml.2012.12.001

Arciuli, J. (2018). Reading as statistical learning. *Language, Speech, and Hearing Services in Schools* 49, 634–643. doi:10.1044/2018_LSHSS-STLT1-17-0135

Bates, D., Mächler, M., Bolker, B., and Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* 67, 1–48. doi:10.18637/jss.v067.i01

Bertram, R., Baayen, H., and Schreuder, R. (2000). Effects of family size for complex words. *Journal of Memory and Language* 42, 390–405. doi:10.1006/jmla.1999.2681

Beyersmann, E., Cavalli, E., Casalis, S., and Colé, P. (2016). Embedded Stem Priming Effects in Prefixed and Suffixed Pseudowords. *Scientific Studies of Reading* 20, 220–230. doi:10.1080/10888438.2016.1140769

Bolger, D. J., Balass, M., Landen, E., and Perfetti, C. A. (2008). Context Variation and Definitions in Learning the Meanings of Words: An Instance-Based Learning Approach. *Discourse Processes* 45, 122–159. doi:10.1080/01638530701792826

Brown, J. I., Bennett, J. M., and Hanna, G. S. (1993). *The Nelson-Denny Reading Test: Vocabulary, Comprehension, Reading Rate. Forms E and F* (Riverside)

Bryant, P., Deacon, H., Nunes, T., Joshi, R. M., and Aaron, P. G. (2005). Morphology and spelling: What have morphemes to do with spelling? In *Handbook of Orthography and Literacy* (New York, NY, US: Routledge). 601–616

Castles, A. and Nation, K. (2006). How does orthographic learning happen? In *From Inkmarks to Ideas*, ed. S. Andrews (London: Psychology Press). 181–209

Chuang, Y.-Y. and Baayen, R. H. (2021). Discriminative Learning and the Lexicon: NDL and LDL. In *Oxford Research Encyclopedia of Linguistics*, ed. M. Aronoff (Oxford University Press). 0. doi:10.1093/acrefore/9780199384655.013.375

de Jong, N. H., Feldman, L. B., Schreuder, R., Pastizzo, M. J., and Baayen, H. (2002). The processing and representation of Dutch and English compounds: Peripheral morphological and central orthographic effects. *Brain and Language* 81, 555–567. doi:10.1006/brln.2001.2547

de Jong, N. H., Schreuder, R., and Baayen, H. (2000). The morphological family size effect and morphology. *Language and Cognitive Processes* 15, 329–365. doi:10.1080/01690960050119625

Ford, M. A., Davis, M. H., and Marslen-Wilson, W. D. (2010). Derivational morphology and base morpheme frequency. *Journal of Memory and Language* 63, 117–130. doi:10.1016/j.jml.2009.01.003

Grainger, J. and Holcomb, P. J. (2009). Watching the word go by: On the time-course of component processes in visual word recognition. *Language and linguistics compass* 3, 128–156

Grainger, J., Kiyonaga, K., and Holcomb, P. J. (2006). The time course of orthographic and phonological code activation. *Psychological Science* 17, 1021–1026

Holcomb, P. J. and Grainger, J. (2006). On the time course of visual word recognition: An event-related potential investigation using masked repetition priming. *Journal of Cognitive Neuroscience* 18, 1631–1643

Holcomb, P. J. and Grainger, J. (2007). Exploring the temporal dynamics of visual word recognition in the masked repetition priming paradigm using event-related potentials. *Brain Research* 1180, 39

Kiyonaga, K., Grainger, J., Midgley, K., and Holcomb, P. J. (2007). Masked cross-modal repetition priming: An event-related potential investigation. *Language and Cognitive Processes* 22, 337–376

Kutas, M. and Federmeier, K. D. (2011). Thirty years and counting: Finding meaning in the N400 component of the event related brain potential (ERP). *Annual review of psychology* 62, 621–647. doi:10.1146/annurev.psych.093008.131123

Kutas, M. and Hillyard, S. A. (1980). Reading senseless sentences: Brain potentials reflect semantic incongruity. *Science* 207, 203–205

Kuznetsova, A., Brockhoff, P. B., and Christensen, R. H. B. (2017). lmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software* 82, 1–26. doi:10.18637/jss.v082.i13

Lenth, R. V. and Piaskowski, J. (2026). *Emmeans: Estimated Marginal Means, Aka Least-Squares Means*

Li, J., Behzadnia, A., and Beyersmann, E. (2026). Learning novel complex words in a second language: Effects of morpheme training and family size. *Studies in Second Language Acquisition* 48, 329–349. doi:10.1017/S0272263125101393

Longtin, C.-M. and Meunier, F. (2005). Morphological decomposition in early visual word processing. *Journal of Memory and Language* 53, 26–41

Morris, J., Frank, T., Grainger, J., and Holcomb, P. J. (2007). Semantic transparency and masked morphological priming: An ERP investigation. *Psychophysiology* 44, 506–521. doi:10.1111/j.1469-8986.2007.00538.x

Perfetti, C. A. (2007). Reading ability: Lexical quality to comprehension. *Scientific Studies of Reading* 11, 357–383. doi:10.1080/10888430701530730

Perfetti, C. A. (2017). Lexical quality revisited. In *Developmental Perspectives in Written Language and Literacy: In Honor of Ludo Verhoeven*, eds. E. Segers and P. v. den Broek (Amsterdam, Netherlands: John Benjamins). 51–67

Perfetti, C. A. and Hart, L. (2002). The lexical quality hypothesis. In *Precursors of Functional Literacy*, eds. L. Verhoeven, C. Elbro, and P. Reitsma (John Benjamins Publishing), vol. 11. 67–86

R Core Team (2026). *R: A Language and Environment for Statistical Computing*. Vienna, Austria

Rastle, K. and Davis, M. H. (2008). Morphological decomposition based on the analysis of orthography. *Language and Cognitive Processes* 23, 942–971

Rayner, K., Foorman, B. R., Perfetti, C. A., Pesetsky, D., and Seidenberg, M. S. (2001). How psychological science informs the teaching of reading. *Psychological Science in the Public Interest* 2, 31–74. doi:10.1111/1529-1006.00004

Sawi, O. (2016). *Individual Differences in Sensitivity to Morphological Structure in Words and Nonwords*. Ph.D. thesis

Schreuder, R. and Baayen, H. (1995). Modeling morphological processing. In *Morphological Aspects of Language Processing.*, ed. L. B. Feldman (Hillsdale, NJ England: Lawrence Erlbaum Associates, Inc). 131–154

Schreuder, R. and Baayen, H. (1997). How complex simplex words can be. *Journal of Memory and Language* 37, 118–139. doi:10.1006/jmla.1997.2510

Stanovich, K. E. and West, R. F. (1989). Exposure to Print and Orthographic Processing. *Reading Research Quarterly* 24, 402–433

Taft, M. (1979). Recognition of affixed words and the word frequency effect. *Memory & Cognition* 7, 263–272

Taft, M. (2004). Morphological decomposition and the reverse base frequency effect. *The Quarterly Journal of Experimental Psychology A: Human Experimental Psychology* 57A, 745–765

Taft, M. and Forster, K. I. (1975). Lexical storage and retrieval of prefixed words. *Journal of Verbal Learning & Verbal Behavior* 14, 638–647

Treiman, R. (2018). Statistical learning and spelling. *Language, Speech, and Hearing Services in Schools* 49, 644–652. doi:10.1044/2018_LSHSS-STLT1-17-0122

Ziegler, J. and Goswami, U. (2005). Reading acquisition, developmental dyslexia, and skilled reading across languages: A psycholinguistic grain size theory. *Psychological Bulletin* 131, 3–29

Supplementary Materials

Reading skill and the time course of morphological processing: electrophysiological evidence for stage-specific modulation

Overview

This document reports full model output for analyses summarised briefly in the main text. Sections S1–S2 report the word morphological family size analyses. In the main manuscript, base frequency is treated as the primary morphological predictor for word ERP analyses, since the study's central claim — that Orthographic Sensitivity (OS) and Language Proficiency (LP) show distinct, component-specific patterns of modulation, with OS the primary modulator at the N250 and LP the primary modulator at the N400 — is formally tested using base frequency as the morphological variable of interest. Family size analyses for words were conducted in parallel, using the same model structure, and are summarised briefly in the main text (see *Words N250: Base Frequency* and *Words N400: Base Frequency*). Sections S3–S4 report exploratory models that reverse the interaction structure at each time window, testing whether the non-predicted individual difference dimension — Language Proficiency at the N250, Orthographic Sensitivity at the N400 — also modulated processing of base frequency and family size, with the predicted dimension included as a covariate. The full statistical output for all four exploratory models, along with the two family size models, is reported below for transparency and to allow readers to evaluate each pattern in full.

S1. Words N250: Family Size

Mean N250 amplitudes for word stimuli were analyzed using a linear mixed-effects model with fixed effects of Family Size, Orthographic Sensitivity (OS), Language Proficiency (LP), and the interaction between Family Size and OS. The model included random intercepts for participants and for electrode site nested within participant. The analysis revealed a robust main effect of Family Size—words from smaller morphological families elicited more negative N250 amplitudes than words from larger families, $M_{\text{large}} = -0.57\,\mu\text{V}$, 95% CI $[-1.03, -0.11]$, $M_{\text{small}} = -0.90\,\mu\text{V}$, 95% CI $[-1.36, -0.44]$, $\beta = 0.324$, $SE = 0.058$, $t(1618) = 5.54$, $p < .001$, $\eta_p^2 = .02$. Neither Orthographic Sensitivity, $\beta = 0.261$, $SE = 0.256$, $t(57) = 1.02$, $p = .31$, nor Language Proficiency, $\beta = -0.279$, $SE = 0.187$, $t(57) = -1.50$, $p = .14$, yielded reliable main effects. The interaction between Family Size and Orthographic Sensitivity did not reach significance but showed a marginal trend, $\beta = -0.111$, $SE = 0.065$, $t(1618) = -1.70$, $p = .089$, $\eta_p^2 = .002$, suggesting that the family size effect at the N250 may be modulated by orthographic skill, though this requires cautious interpretation given the marginal status of the effect. The model explained 1.9% of the variance in N250 amplitude via fixed effects ($R^2_{\text{marginal}} = .019$) and 73.8% overall ($R^2_{\text{conditional}} = .738$), with the large conditional R^2 reflecting substantial variability in N250 amplitude across participants and electrode sites. Together, these results indicate that morphological family structure robustly modulates early morpho-orthographic processing for words, with limited but suggestive evidence that orthographic skill may shape the sensitivity of this effect.

Table 1: Words N250: Family Size model, fixed effects.

Term	β	**SE**	**df**	**t**	**p**
Family Size	0.324	0.058	1618	5.54	$< .001$
Orthographic Sensitivity	0.261	0.256	57	1.02	.31
Language Proficiency	−0.279	0.187	57	−1.50	.14
Family Size × OS	−0.111	0.065	1618	−1.70	.089

$R^2_{\text{marginal}} = .019$; $R^2_{\text{conditional}} = .738$. Random effects: participant, electrode nested within participant.

S2. Words N400: Family Size

Mean N400 amplitudes for word stimuli were analyzed using a linear mixed-effects model with fixed effects of Family Size, Language Proficiency (LP), Orthographic Sensitivity (OS), and the interaction between Family Size and LP. The model included random intercepts for participants and for electrode site nested within participant. The analysis revealed a robust main effect of Family Size—words from smaller morphological families elicited greater N400 amplitudes than words from larger families, $M_{\text{large}} = 0.518\,\mu\text{V}$, 95% CI $[-0.054, 1.089]$, $M_{\text{small}} = 0.062\,\mu\text{V}$, 95% CI $[-0.509, 0.634]$, $\beta = 0.452$, $SE = 0.064$, $t(1618) = 7.07$, $p < .001$, $\eta_p^2 = .03$, indicating facilitated semantic integration for words supported by richer morphological families. A significant main effect of Orthographic Sensitivity was also observed, $\beta = 0.677$, $SE = 0.318$, $t(57) = 2.13$, $p = .038$, $\eta_p^2 = .07$, reflecting overall differences in N400 amplitude as a function of orthographic skill independently of morphological structure. Language Proficiency did not yield a reliable main effect, $\beta = 0.001$, $SE = 0.231$, $t(57) = 0.005$, $p = .996$, nor did the Family Size × LP interaction reach significance, $\beta = -0.071$, $SE = 0.052$, $t(1618) = -1.37$, $p = .172$. These results indicate that semantic integration of morphological family information for familiar words is robust and not selectively modulated by individual differences in language proficiency.

Table 2: Words N400: Family Size model, fixed effects.

Term	β	**SE**	**df**	**t**	**p**
Family Size	0.452	0.064	1618	7.07	$< .001$
Language Proficiency	0.001	0.231	57	0.005	.996
Orthographic Sensitivity	0.677	0.318	57	2.13	.038
Family Size × LP	−0.071	0.052	1618	−1.37	.172

S3. Words N250: Exploratory Language Proficiency Models

To assess whether Language Proficiency also modulated the N250 base frequency effect for words, an exploratory model was fit with fixed effects of Base Frequency, Language Proficiency (LP), Orthographic Sensitivity (OS), and the interaction between Base Frequency and LP; this reverses the structure of the primary model reported in the main text, in which OS rather than LP was the interacting term. The model included random intercepts for participants and for electrode site nested within participant. The main effect of Base Frequency was significant, $\beta = -0.329$, $SE = 0.100$, $t(1618) = -3.27$, $p = .001$, $\eta_p^2 = .007$, consistent with the primary model. Neither Language Proficiency, $\beta = -0.536$, $SE = 0.378$, $t(57) = -1.42$, $p = .162$, nor Orthographic Sensitivity, $\beta = 0.534$, $SE = 0.520$, $t(57) = 1.03$, $p = .309$, yielded reliable main effects. Critically, the interaction between Base Frequency and Language Proficiency was significant, $\beta = 0.180$, $SE = 0.082$, $t(1618) = 2.20$, $p = .028$, $\eta_p^2 = .003$. As reported in the main text, follow-up contrasts indicated that this modulation ran in the opposite direction to the Orthographic Sensitivity interaction in the primary model: the base frequency effect was largest at low LP and attenuated steadily as LP increased, becoming non-significant by LP = +1 SD ($\Delta = -0.15\,\mu$V, $t(1618) = -1.13$, $p = .259$) and reversing in direction, though not significantly, at the upper end of the observed range (LP = +3 SD: $\Delta = 0.21\,\mu$V, $t(1618) = 0.78$, $p = .433$). The model explained 1.8% of variance via fixed effects ($R^2_{\text{marginal}} = .018$) and 79.9% overall ($R^2_{\text{conditional}} = .799$).

Table 3: Words N250, Base Frequency × LP (exploratory), fixed effects.

Term	β	**SE**	**df**	**t**	**p**
Base Frequency	−0.329	0.100	1618	−3.27	.001
Language Proficiency	−0.536	0.378	57	−1.42	.162
Orthographic Sensitivity	0.534	0.520	57	1.03	.309
Base Frequency × LP	0.180	0.082	1618	2.20	.028

$R^2_{\text{marginal}} = .018$; $R^2_{\text{conditional}} = .799$. Random effects: participant, electrode nested within participant.

A parallel exploratory model tested the interaction between Family Size and Language Proficiency at the N250, with fixed effects of Family Size, Language Proficiency (LP), Orthographic Sensitivity (OS), and the interaction between Family Size and LP; Orthographic Sensitivity was again included as a covariate. The main effect of Family Size was significant, $\beta = 0.325$, $SE = 0.058$, $t(1618) = 5.56$, $p < .001$, $\eta_p^2 = .02$, consistent with the primary family size model reported in Table S1. Neither Language Proficiency, $\beta = -0.279$, $SE = 0.186$, $t(57) = -1.50$, $p = .140$, nor Orthographic Sensitivity, $\beta = 0.261$, $SE = 0.256$, $t(57) = 1.02$, $p = .313$, yielded reliable main effects. The interaction between Family Size and Language Proficiency was not significant, $\beta = -0.074$, $SE = 0.047$, $t(1618) = -1.56$, $p = .119$, indicating that, unlike base frequency, the family size effect at the N250 was not modulated by Language Proficiency. The model explained 1.9% of variance via fixed effects ($R^2_{\text{marginal}} = .019$) and 73.8% overall ($R^2_{\text{conditional}} = .738$).

Table 4: Words N250, Family Size × LP (exploratory), fixed effects.

Term	β	**SE**	**df**	**t**	**p**
Family Size	0.325	0.058	1618	5.56	< .001
Language Proficiency	−0.279	0.186	57	−1.50	.140
Orthographic Sensitivity	0.261	0.256	57	1.02	.313
Family Size × LP	−0.074	0.047	1618	−1.56	.119

$R^2_{\text{marginal}} = .019$; $R^2_{\text{conditional}} = .738$. Random effects: participant, electrode nested within participant.

S4. Words N400: Exploratory Orthographic Sensitivity Models

To assess whether Orthographic Sensitivity also modulated the N400 base frequency effect for words, an exploratory model was fit with fixed effects of Base Frequency, Orthographic Sensitivity (OS), Language Proficiency (LP), and the interaction between Base Frequency and OS; this reverses the structure of the primary model reported in the main text, in which LP rather than OS was the interacting term. The model included random intercepts for participants and for electrode site nested within participant. The main effect of Base Frequency showed a marginal trend, $\beta = 0.189$, $SE = 0.107$, $t(1618) = 1.77$, $p = .077$, $\eta_p^2 = .002$. Orthographic Sensitivity showed a significant main effect, $\beta = 1.290$, $SE = 0.640$, $t(57) = 2.02$, $p = .049$, $\eta_p^2 = .07$, consistent with the OS main effect reported in the primary model. Language Proficiency did not yield a reliable main effect, $\beta = 0.047$, $SE = 0.465$, $t(57) = 0.10$, $p = .920$. The interaction between Base Frequency and Orthographic Sensitivity was not significant, $\beta = 0.195$, $SE = 0.120$, $t(1618) = 1.63$, $p = .104$, indicating that Orthographic Sensitivity did not modulate the N400 base frequency effect. The model explained 2.6% of variance via fixed effects ($R^2_{\text{marginal}} = .026$) and 82.4% overall ($R^2_{\text{conditional}} = .824$).

Table 5: Words N400, Base Frequency × OS (exploratory), fixed effects.

Term	β	**SE**	**df**	**t**	**p**
Base Frequency	0.189	0.107	1618	1.77	.077
Orthographic Sensitivity	1.290	0.640	57	2.02	.049
Language Proficiency	0.047	0.465	57	0.10	.920
Base Frequency × OS	0.195	0.120	1618	1.63	.104

$R^2_{\text{marginal}} = .026$; $R^2_{\text{conditional}} = .824$. Random effects: participant, electrode nested within participant.

A parallel exploratory model tested the interaction between Family Size and Orthographic Sensitivity at the N400, with fixed effects of Family Size, Orthographic Sensitivity (OS), Language Proficiency (LP), and the interaction between Family Size and OS. The main effect of Family Size was significant, $\beta = 0.452$, $SE = 0.064$, $t(1618) = 7.07$, $p < .001$, $\eta_p^2 = .03$, consistent with the primary family size model reported in Table S2. Orthographic Sensitivity showed a significant main effect, $\beta = 0.677$, $SE = 0.318$, $t(57) = 2.13$, $p = .038$, $\eta_p^2 = .07$, consistent with the OS main effect reported in Table S2. Language Proficiency did not yield a reliable main effect, $\beta = 0.001$, $SE = 0.231$, $t(57) = 0.01$, $p = .996$. The interaction between Family Size and Orthographic Sensitivity was not significant, $\beta = 0.091$, $SE = 0.072$, $t(1618) = 1.27$, $p = .204$, indicating that, unlike Language

Proficiency, Orthographic Sensitivity did not modulate the N400 family size effect. The model explained 3.1% of variance via fixed effects ($R^2_{\text{marginal}} = .031$) and 75.6% overall ($R^2_{\text{conditional}} = .756$).

Table 6: Words N400, Family Size × OS (exploratory), fixed effects.

Term	β	**SE**	**df**	**t**	**p**
Family Size	0.452	0.064	1618	7.07	$< .001$
Orthographic Sensitivity	0.677	0.318	57	2.13	.038
Language Proficiency	0.001	0.231	57	0.01	.996
Family Size × OS	0.091	0.072	1618	1.27	.204

$R^2_{\text{marginal}} = .031$; $R^2_{\text{conditional}} = .756$. Random effects: participant, electrode nested within participant.

References